

FORM 3 -

MATHEMATICS

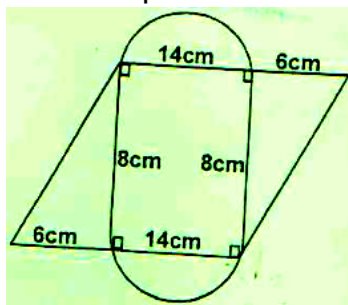
- Paper 1/2

EXAM 1
PAPER 1

Section I (50 marks)

Answer all the questions in this section

1. A basketball team play 10 matches in a tournament. The following are scores in each match.
9, 15, 17, 16, 7, 20, 21, 15, 10, 1
Determine:
(a) the mode. (1 mark)
(b) the median. (1 marks)
2. The coordinates of P and Q are (-2, 6) and (4, -2) respectively. Find the equation of a perpendicular bisector of line PQ, in the form $y = mx + c$. (4 marks)
3. The marked price of a car in a dealer's shop was Ksh. 450,000/=. Magari bought the car at 7% discount. The dealer still made a profit of 13%. Calculate the amount of money the dealer had paid for the car. (3marks)
4. Simplify $\frac{12x^2 + ax - 6a^2}{9x^2 - 4a^2}$. (3marks)
5. Solve for m in the equation:
 $3^{4(m+1)} + 3^{4m} = 246$ (3marks)
6. Using tables evaluate.
 $\frac{1}{34.52} + \sqrt[3]{0.787} + (0.934)^3$ (3 marks)
7. Evaluate without using a calculator $\frac{\frac{2}{3} \left(1\frac{3}{7} - \frac{5}{8} \right)}{\frac{3}{4} + 1\frac{5}{7} \div \frac{4}{7} \text{ of } 2\frac{1}{3}}$. (3marks)
8. The exterior angle of a regular polygon is 24° . Determine the sum of the interior angles. (3marks)
9. The figure below represents an opened collar cloth, find the distance round it. (Take $\pi = 3\frac{1}{7}$) (3marks)



10. An American tourist arrives in Kenya with 1000 US\$ and converted the whole amount into Kenyan shilling. He spent sh. 40000 and changed the balance to Sterling pounds before leaving for United Kingdom. A Kenyan bank buys and sells foreign currencies as shown.

	Buying (in Kshs)	Selling (in ksh)
1 US dollar	84.2083	84.3806
1 Sterling pound	134.7941	135.1294

Calculate the amount he received to the nearest sterling pound.

(3 marks)

11. Katu is now four times as old as her daughter and six times as old as her son. Twelve years from now, the sum of the ages of her daughter and son will differ from her age by 9 years. What is Katu's present age?

(3 marks)

12. Solve the following inequality and show your solution on a number line.

(3 marks)

$$4x - 3 \leq \frac{1}{2}(x + 8) < x + 5$$

13. A farmer has a piece of land measuring 840m by 396m. He divides it into square plots of equal size. Find the maximum area of one plot.

(3marks)

14. Using a ruler and a pair of compasses only, construct a triangle ABC in which BC = 5cm, angle ABC = 75° and ACB = 60° . From A drop a perpendicular to BC and measure its length to the nearest mm.

(4 marks)

15. A two digit number is such that the sum of the digits is 11. When the digits are reversed, the new number exceeds the original number by 9. Calculate the original number.

(4marks)

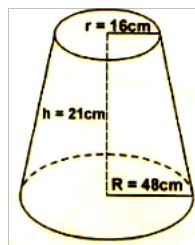
16. Two similar containers have masses 256kgs and 128kgs respectively. The surface area of the smaller container is 810 cm^2 . What is the area of the corresponding surface of the large container?

(3marks)

SECTION II (50 MARKS)

Answer any FIVE questions in this section

17. The figure below is a frustum of a solid cone of base radius 48cm and top radius 16cm. The height of the frustum is 21cm.



Taking by as $\frac{22}{7}$, calculate:

- The height of the solid cone.
- The volume of the solid frustum.
- The total surface area of the frustum.

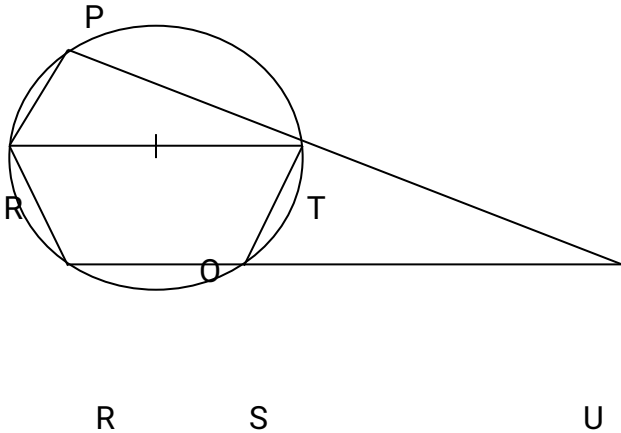
(2marks)

(3marks)

(5marks)

18. The figure below shows a circle centre O in which QOT is a diameter. $\angle QTP = 46^\circ$, $\angle TQR = 75^\circ$ and $\angle SRT = 38^\circ$,

PTU and RSU are straight lines.



Calculate the following angles giving a reason in each case.

- | | | |
|-----|---------------------|----------|
| (a) | $\angle RST$ | (2marks) |
| (b) | $\angle SUT$ | (2marks) |
| (c) | $\angle PST$ | (2marks) |
| (d) | Obtuse $\angle ROT$ | (2marks) |
| (e) | $\angle SQT$ | (2marks) |

19.(a) A rectangular tank of base 2.4m by 2.8m and a height of 3m contains 3600 litres of water initially. Water flows into the tank at the rate of 0.5 litres per second.

Calculate:

- | | | |
|------|--|----------|
| (i) | The amount needed to fill the tank. | (2marks) |
| (ii) | The time in hours and minutes required to fill. | (3marks) |
| b) | Pipe A can fill an empty tank in 3 hours while pipe B can fill the same tank in 6 hours. When the tank is full, it can be emptied by pipe C in 8 hours. Pipes A and B are opened at the same time when the tank is empty. If one hour later pipe C is also opened, find the total time taken to fill the tank. | (5marks) |

20. A salesman is paid a commission of 2% on goods worth over Ksh.100000. He is also paid a monthly salary of Ksh.12000. In a certain month, he sold 360 pairs of shoes at Ksh.500 each pair.

- | | | |
|-----|---|-----------|
| (a) | Calculate the salesman's earning that month. | (3 marks) |
| b) | The following month, his monthly salary was increased by 10%. His total earnings that month were Ksh.17600. | |
| | Calculate | |
| (i) | The total amount of money received from the sales of the shoes that month. | (5 marks) |

(ii) The number of pairs of shoes sold that month. (2 marks)

21. The distance between towns M and N is 280km. A car and a lorry travel from M to N. The average speed of the lorry is 20km/h less than that of the car. The lorry takes 1h 10min more than the car to travel from M to N.

(a) If the speed of the lorry is x km/h, find x (6marks)

(b) The lorry left town M at 8.15am. The car left town M later and overtook the lorry at 12.15pm. Calculate the time the car left town M. (4marks)

22. The table below shows measurements, in metres made by surveyor in his field book.

	F	
	420	
G 100	380	D70
	300	C100
	220	E40
H60	140	
	80	B60
	A	

a) Using an appropriate scale draw the region. (5 marks)

b) Find the area in hectares of the field. (5 marks)

23. A, B, C and D are four schools where B is 84km north of A and C is on a bearing of $N65^{\circ}W$ from A at a distance of 60km. D is on a bearing of $N20^{\circ}W$ from C and at a distance of 30km.

Use a scale drawing to show relative positions of A, B, C and D using a scale of 1cm to represent 10km. (5marks)

Find;

(a) the distance and bearing of B from C. (2marks)

(b) the bearing and distance of D from B. (2marks)

(c) the bearing of A and D. (1mark)

24. A company is to construct a parking bay whose area is $135m^2$. It is to be covered with a concrete slab of uniform thickness of 150mm. To make the slab, cement, ballast and sand are to be mixed so that their masses are in the ratio 1: 4: 4. The mass of $1m^3$ of dry slab is 2500kg. Calculate

a)i) the volume of the slab. (2marks)

ii) the mass of the dry slab. (1 mark)

iii) the mass of cement to be used. (2 marks)

b) If one bag of cement is 50kg, find the number of bags to be purchased. (2 marks)

c) If a lorry carries 7 tonnes of sand, calculate the number of lorries of sand to be purchased. (3 marks)

PAPER 2

Section I (50 marks)

Answer all the questions in this section

1. Solve for x in the equation $\sin(4x - 10)^{\circ} - \cos(x + 60)^{\circ} = 0$ (3 marks)

2. (a) Find the greatest common divisor of the term $9x^3y^2$ and $4xy^4$.

(1mark)

(b) Hence factorize completely the expression $9x^3y^2 - 4xy^4$

(2marks)

3 The area of a rhombus is 120 cm^2 . Given that one of its diagonals is 24 cm , calculate the perimeter of the rhombus.

(3 marks)

4. Given that $\vec{OA} = 2\mathbf{i} + 3\mathbf{j}$ and $\vec{OB} = 3\mathbf{i} - 2\mathbf{j}$. Find the magnitude of $\vec{AB}$ to one decimal place.

(3marks)

5. Given that $\log_{10} 7 = 0.8451$ and $\log_{10} 6 = 0.7782$. find $\log_{10} 25.2$

(4 marks)

6. The angle subtended by the major arc at the centre of the circle O is twice the angle subtended by the minor arc at the centre. If the radius of the circle is 3.5 cm , find the length of the minor arc. (Take $\pi = \frac{22}{7}$)

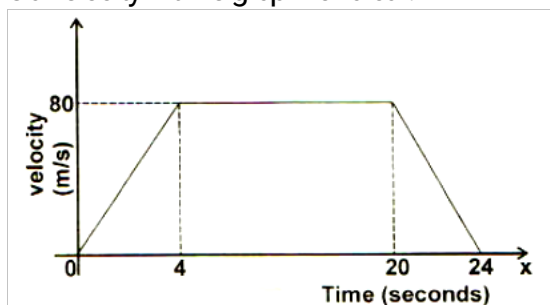
7. Simply and leave answer in surd form.

(4 marks)

$$\frac{-9}{\sqrt{13} + \sqrt{3}} - \frac{5}{\sqrt{3} - \sqrt{13}}$$

8. A line L is perpendicular to $3y - 4x = 7$. Determine the acute angle between L and the x-axis. (3marks)

9. The figure below is a velocity – time graph for a car.



a) Find the total distance travelled by the car.

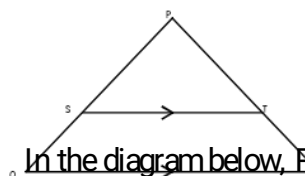
(2marks)

b) Calculate the deceleration of the car.

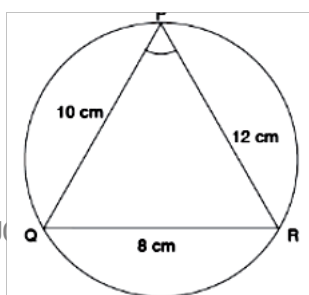
(1marks)

10. The figure below shows triangle PQR in which $PR = 12 \text{ cm}$. T is a point on PR such that $TR = 4 \text{ cm}$. Line ST is parallel to QR. If the area of triangle PQR is 336 cm^2 , find the area of the quadrilateral STQR.

(3 marks)



11. In the diagram below, P, Q and R are points on the circumference of a circle. $PQ = 10 \text{ cm}$, $PR = 12 \text{ cm}$ and $QR = 8 \text{ cm}$



Find the radius of the circle to 2 decimal places.

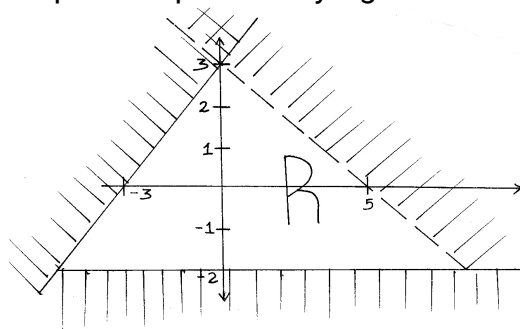
(3 marks)

12. Solve for x in the equation: $2 + \log_7 (3x - 4) = \log_7 98$

(3 mark)

13. Form the inequalities represented by region R.

(3 marks)



14. From the roof of a house, a boy can see an avocado tree which is 20m away from the house. He measures the angle of elevation of the top of the tree as 21° and the angle of depression of the bottom of tree as 31° . Find the height of the avocado tree.

(3 marks)

15. Find the percentage error in calculating the volume of the cuboid whose dimensions are 8.2cm by 6.2cm by 5.7cm

(3 marks)

16. A bus moving at a speed of 80km/h is being overtaken by a car moving at 100km/h in a clear section of a road. Given that the bus is 21m long and the car is 4m long. How much time (in seconds) will elapse before the car can completely overtake the bus?

(3marks)

SECTION II: (50 MARKS)

Answer any five questions from this section

17. (a) Complete the table below for the function $y = \sin 2x^\circ$ and $y = 3 \cos x^\circ$ for $-180^\circ \leq x \leq 180^\circ$.

(2 marks)

x°	-	-	-	-	-	-	0	30	60	90	120	150	180
-----------	---	---	---	---	---	---	---	----	----	----	-----	-----	-----

	1 8 0	1 5 0	1 2 0	9 0	6 0	3 0					0	0
Si n 2 x 0	0			0	- 0 . 8 7				0. 8 7	0		
3C o s x 0	-3	- 2 . 6		0		2. 6					- 1 . 5	

(b) On the same axes, draw the graph of $y = \sin 2x^\circ$ and $y = 3 \cos x^\circ$ for $-180^\circ \leq x \leq 180^\circ$.
(5marks)

(c) Use the graph in (b) above to find:

(i) the value of x such that $3 \cos x^\circ - \sin 2x^\circ = 0$.

(1 mark)

(ii) the difference in value of y when $x = 45^\circ$.

(1 mark)

(iii) Range of values of x such that $3 \cos x > 1.5$.

(1mark)

18. (a) Complete the table below for the equation

$$y = x^2 + 3x - 6, \text{ given } -6 \leq x \leq 4.$$

(2marks)

X	-6	-5	-4	-3	-2	-1	0	1	2	3	4
Y	12			-6			-6				22

(b) Using a scale of 1 cm to represent 1 unit in the x-axis and 2 units in the y-axis, draw the graph of $y = x^2 + 3x - 6$.
(4marks)

(c) Use your graph to solve the quadratic equation.

i) $x^2 + 3x - 6 = 0$

(1mark)

ii) $x^2 + 3x - 2 = 0$

(3marks)

19.a) An enlargement with centre $(-2, 3)$ maps $(1, 0)$ onto $(4, -3)$. What is the image of $(-3, -6)$ with the same centre of enlargement.

(4marks)

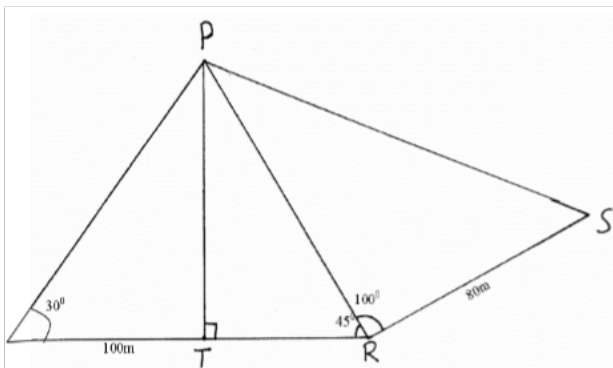
b). If $\frac{9}{98}x^2 - \frac{3}{28}x + \frac{3}{16} + B$ is a perfect square. Find the value of B. (3 marks)

c. If $\cos \theta^\circ = \frac{-15}{17}$ and θ is obtuse, find without using tables the values of

i) $\tan \theta^\circ$. (2marks)

ii) $\sin (180 - \theta)^\circ$ (1mark)

20. The figure below represents a quadrilateral piece of land PQRS divided into three triangular plots. The length QT and RS are 100m and 80m respectively. Angle PQT = 30° , angle PRT = 45° and angle PRS = 100° .



(a) Find to four significant figures

(i) the length of PT. (2marks)

(ii) The length of PS. (3marks)

(iii) the perimeter of the piece of land. (2marks)

(b) The plots are to be fenced with five strands of barbed wire leaving an entrance of 2.8m wide to each plot. The type of marked wire to be used is sold in rolls of length of 480m. Calculate the number of rolls of marked wire that must be bought to complete the fencing of the plot. (3marks)

21. Musa obtained a loan in which the rate of interest was charged at 3.5% quarterly. He cleared this loan by paying Ksh: 24,805 at the end of $1\frac{1}{2}$ years.

(a) How much was the loan if the rate was on:

(i) Simple interest. (4 marks)

(ii) Compound interest. (4 marks)

(b) Calculate the difference between simple and compound interest. (2 marks)

22. Two circles, centre A and B, have radii 4cm and 8cm respectively. If the two circles share a common chord 6cm long, (Taking $\pi = 3.142$).

(a) Calculate the area of the common region. (6 marks)

(b) After constructing the two circles, each separately on a piece of paper, a student cut out a minor segment on each circle along the chord 6cm and joined the major segments along the chords. Find the perimeter of the figure made. (4 marks)

23. The following data was obtained for masses of some pregnant women in a maternity clinic

Mass x (kg)	$1.5 \leq x < 5.5$	$5.5 \leq x < 7.5$	$7.5 \leq x < 13.$ 5	$13.5 \leq x < 15.$ 5	$15.5 \leq x < 20.$ 5
No. Of women	16	20	18	14	15

(a) Represent the information in the table above on a histogram, on the graph paper provided. (6marks)

(b) Use the information in the table above to estimate:

(i) the mean mass. (2marks)

(ii) the median mass.

24. a) A room is constructed such that its external length and breadth are 7.5m and 5.3m respectively. The thickness of the wall is 15cm and its height is 3.3 metres. A total space of 5 m^2 is left for doors and windows on the walls.

a) Calculate the volume of:

i) the materials needed to construct the walls with the doors and windows. (4 marks)

ii) the materials needed to construct the walls without doors and windows.

2 marks)

b) The blocks used in constructing the walls are 450mm by 200mm by 150mm. 0.225m³ of cement is used to join the blocks. Calculate the number of blocks. Calculate the number of blocks needed to construct the room

(4 marks)

EXAM 2

PAPER 1

1. Without using mathematical tables or calculators, evaluate the following leaving your answer as a fraction in its simplest form (3mks)

$$^3\sqrt{\frac{0.119 \times 0.256}{0.068 \times 7}}$$

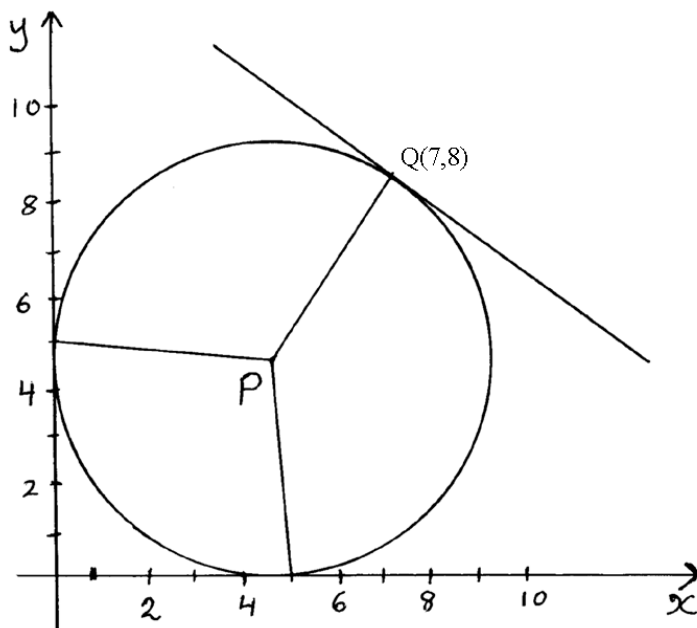
2. Two boys and a girl shared some money .The elder boy got $\frac{4}{5}$ of it, the younger boy got $\frac{3}{5}$ of the remainder and the girl got the rest. Find the percentage share of the younger boy to the girl's share.

(4mks)

3. From a point A, the angle of elevation of top of a watch tower is 20°.From another point which is 25m from the base of the tower, the angle of elevation of the top of the tower is 26°.Giving your answer to three decimal places, determine the height of the tower and hence calculate the distance between the points A and B if they are both on the same side of the tower and lie on a straight line with the base of the tower.

(3mks)

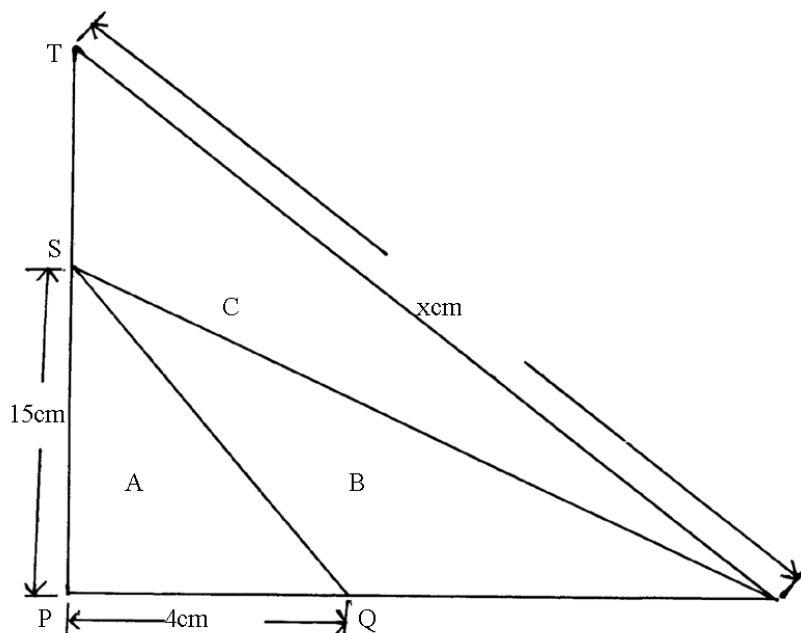
4. If X is a positive integer, find all possible values of x given that $1 < \frac{2}{5}x^2 < 7$. (3mks)
5. A train whose length is 60 metres is travelling at 40km/h in the same direction as a bus whose length is 20m. If the speed of bus is 80km/h and moving parallel to the train, calculate the time it takes the truck to overtake the train completely in seconds. (3mks)
6. A positive two digit number is such that the product of the digits is 20. When the digits are reversed, the number so formed is greater than the original number by 9. Find the number.
7. The diagram below shows a circle with centre $P(5,5)$ and radius 5 units



- (a) Write down in terms of x and y the equation of the circle in the form $ax^2+by^2+cx+dy+e=0$ where a, b, c, d and e are constants. (1mk)
- (b) Determine the gradient of PQ (1mk)

(c) Find the equation of the tangent at Q in the form $ax+by=c$. (2mks)

8. Find the value of x in the following figure given that $\text{area A} = \text{Area B} = \text{Area C}$ (Give your answer to 2 decimal places) (4mks)



9. Odhis car Hire company hires out as follows; sh. 2500 per day and sh. 270 per kilometer covered. They offer a discount of 30km free each day of hire. Makori hires a car for 5 days and drives for 480km. Calculate the total cost. (2mks)

10. Omwando borrows sh. 90,000 for 5 years at $6\frac{1}{2}\%$ simple interest p.a. What amount does he have to pay at the end of that time? (3mks)

11. Solve for t in the equation

$$9^{t+1} + 3^{2t} = 30.$$

(3mks)

12. Given the curve $y = x^2 - 2x + 6$, find the coordinates of the point on the curve at which the gradient is 4.

13. Mary has some money in two denominations only. Fifty shilling notes and twenty shilling coins. She has three times as many fifty shilling notes as twenty shilling coins. If altogether she has sh. 3400, find the number of fifty shilling notes and 20 shilling coins.

14. In Ngamongo village, a piece of work can be completed by 45 workers in 10 days. They worked for 4 days after which 15 workers were laid off. How many days would it take the remaining workers to complete the work?

(3mks)

15. The table below shows marks obtained by a form three class in a certain school.

Marks (x)	$8 \leq x < 9$	$9 \leq x < 11$	$11 \leq x < 13$	$13 \leq x < 16$	$16 \leq x < 20$	$20 \leq x < 21$
No. of students (y)	2	6	8	3	2	1

Use the table to represent the information on a histogram.

(3mks)

16. Find the inverse of the matrix $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$ and hence solve the simultaneous equations below. (4mks)

$$2x + y = 21$$

$$3x + 2y = 34$$

SECTION II (50 MARKS)

ANSWER ANY FIVE QUESTIONS.

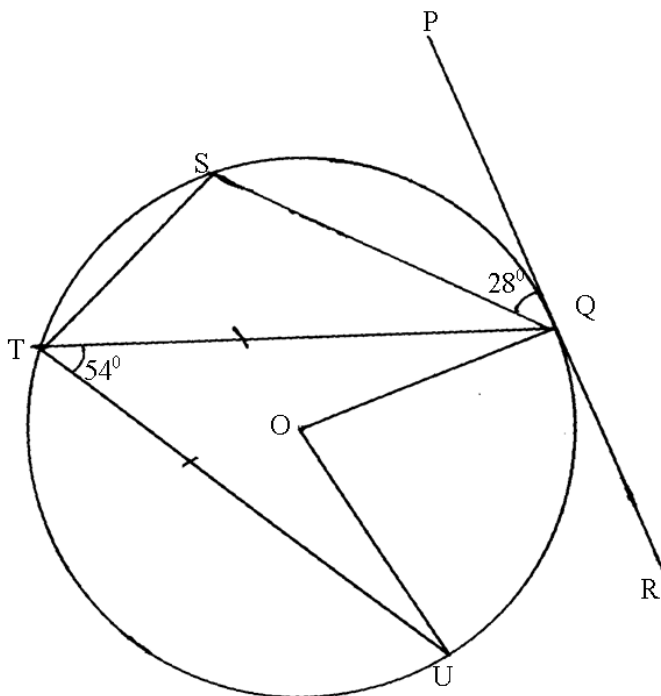
17. A bag contains 5 red, 4 white and 3 blue beads. Two beads are selected at random.

(a) Draw a tree diagram and list the probability space.

(3mks)

- (b) Find the probability that
- (i) The last bead selected is red. (2mks)
 - (ii) The beads selected were of the same colour (2mks)
 - (iii) At least one of the selected beads is blue. (3mks)

18. In the figure below, O is the centre of the circle. PQR is a tangent to the circle at Q . Angle $PQS = 28^\circ$, angle $UTQ = 54^\circ$ and $UT = TQ$.



Giving reasons, determine the size of

(a) Angle STQ. (2mks)

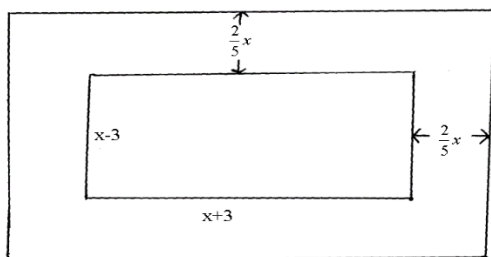
(b) Angle TQU. (2mks)

(c) Angle TQS (2mks)

(d) Reflex angle UOQ. (2mks)

(e) Angle TQR. (2mks)

19. The following figure represent a dancing floor with a carpeted margin all around of $\frac{2}{5}x$ wide leaving a dancing space of $(x-3)m$ by $(x+3)m$



If the total area of the entire room is $315m^2$

(a) Calculate the value of x . (5mks)

(b) Calculate the area of the carpeted margin. (3mks)

(c) If the carpet cost sh. 750 per m^2 , calculate the total cost of the sealed margin. (2mks)

20. John bought 3 brands of tea, A B and C. The cost price of the three brands were sh 25, sh 30, sh 45 per kg respectively. He mixed the three brands in the ratio 5:2:1 respectively. After selling the mixture he made a profit of 20%.

(a) How much profit did he make per kilogram of the mixture? (4 mks)

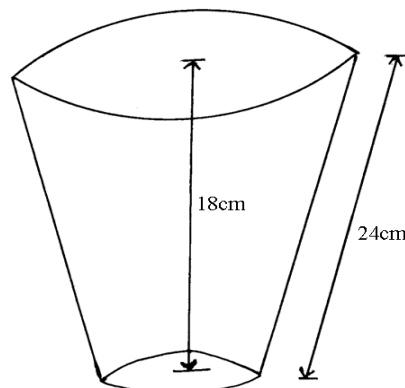
- (b) After one year the cost price of each brand was increased by 10%
- (i) For how much did he sell one kilogram of the mixture to make a profit of 15%? (Give your answer to the nearest 5 cents) (3 mks)
- (ii) What would have been his percentage profit if he sold one kilogram of the mixture at sh. 45.

21. A car accelerates from rest for 10 seconds until it reaches a velocity of 12 metres per second. It then continues at this velocity for the next 40 seconds after which it brakes and comes to rest until a constant retardation of 1.5 metres per second

- (a) Determine
- (i) The acceleration over the first 10 seconds (2mks)
- (ii) The time taken during the retardation (2mks)
- (b) Draw the velocity time graph for the journey and use it to determine.
- (i) The total distance covered by the car (4mks)
- (ii) The percentage of the total distance which was covered during the first 15 seconds. (2mks)

22. The diagram below shows a flower vase of depth 18cm. The ratio of the top and bottom diameters is 5:2 (Take $\pi = 3.142$)

Calculate



(a) The volume of the flower vase (7mks)

(b) The curved surface area of the flower vase (3mks)

23. Given that $x-y=3$ and $3x+y=17$, find without solving for X and Y the value of

(a) $2xy-x^2-y^2$ (2mks)

(b) $6xy+y^2+9x^2$ (2mks)

(c) $3x^2-2xy+y^2$ (3mks)

(d) $\frac{3x^2 - 4xy + y^2}{9x^2 - y^2}$ (3mks)

24. Three Kenyan warships A,B and C are at sea such that ship B is 450km on a bearing of 030° from ship A. ship C is 700km from ship B on a bearing of 120° . An enemy ship D is sighted 1000km due south of ship B.

(a) Taking a scale of 1cm to represent 100km locate the position of the ships A,B,C and D (4mks)

(b) Find the compass bearing of :

(i) Ship A from ship D (1mk)

(ii) Ship D from ship C (1mk)

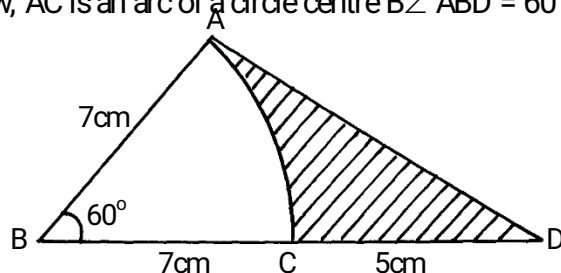
- (c) Use the scale drawing to determine
- (i) The distance of D from A (1mk)
- (ii) The distance of C from D (1mk)
- (d) Find the bearing of :
- (i) B from C (1mk)
- (ii) A from C (1mk)

PAPER 2

1. Evaluate without using tables or calculators. (3mks)
- $$\sqrt{\frac{0.8064 \times 6.048}{1.008 \times 0.1344}}$$
2. Evaluate -4 of $\frac{[(-4 + -5 \div 15) + -3 - 4 \div 6]}{84 \div -7 + 3 - -5}$ (2mks)
3. Solve for θ without using table given that $0 \leq \theta \leq 90^\circ$ and that $\sin (2\theta - 30^\circ) - \cos 4\theta = 0$ (3mks)
4. Solve for x given that $5^{2x+2} - 20 \times 5^{2x} = 625$ (3mks)
5. The angle of a quadrilateral ABCD in order are $2(x-10)$, $4(x+5)$, $5(x+4)$ and $(x-20)$ in degrees. Find the exterior angles of the quadrilateral. (4mks)

6. A radio costing kshs. 1240 is marked to sell at a price calculated to give a profit of 40 %. What will be its selling price in sale when 25% is taken off the marked price? (3mks)
7. Show that if $OA = -i + 7j$, $OB = 3i - 5j$ and $OC = 4j$, then points A, B and C are collinear. (4mks)
8. Four men can dig 2 acres of land in 3 days working 4 hour a day. How many men are required to dig 5 acres of land in 4 days working 3 hours a day at the same rate. (3mks)
9. The surface area of two similar bottles are 12 cm^2 and 108 cm^2 respectively. If the larger one has a volume of 810 cm^3 . Find the volume of the smaller one. (3mks)

10. In the figure given below, AC is an arc of a circle centre B $\angle ABD = 60^\circ$, $AB = BC = 7 \text{ cm}$ and $CD = 5 \text{ cm}$.



Calculate

- a) The area of triangle ADB (2mks)
- b) The area of the shaded region. (2mks)
11. Solve the inequalities and represent the information on the number line. (3mks)
- $$-3 + 2x < 3x + 2 < 4(x - 5)$$

12. Make x the subject of the formula in $3s = 2p\sqrt{\frac{x}{3x-5}}$ (3mks)
13. Given $x = 13.4\text{cm}$ and $y = 4.3\text{ cm}$. calculate the percentage error in $\frac{x}{y}$ correct to 4 d.p(3mks)
14. A straight line through the point A (2, 1) and B (4,m) is perpendicular to the line whose equation is $3y = 5 - 2x$, Determine the value of m (3mks)
15. Okoth deposited some money at 10% compound interest compounded annually. How long will it take to double the amount to the nearest year? (3mks)
16. Chebet has 5 brown chicken and 3 black ones. She picks two of them for slaughter at random, one after the other. What is the probability that the two are of different colours. (3mks)

SECTION II

Answer only five questions

17. A bus left Nairobi at 8.00am and traveled towards Busia at an average speed of 80km/hr. At 8.30 am a car left Busia for Nairobi at an average speed of 120km/hr. Given that the distance between Nairobi and Busia is 400km.

Calculate:

- The time the car arrived in Nairobi. (2mks)
 - The time the two vehicles met. (4mks)
 - The distance from Nairobi to the meeting point. (2mks)
 - The distance of the bus from Busia when the car arrived in Nairobi. (2mks)
18. A triangle whose vertices are A (1,4) B (2,1) and C (5,2) is given the following transformation:
- Reflection in the line $y = -x$ to $A^1B^1C^1$

ii) $A^1B^1C^1$ is then given rotation of $+90^\circ$ about the origin to $A^{11}B^{11}C^{11}$

iii) $A^{11}B^{11}C^{11}$ is then given a translation vector $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$ to $A^{111}B^{111}C^{111}$

iv) $A^{111}B^{111}C^{111}$ is then given an enlargement scale factor -2 centre $(0, 0)$ to $A^{IV}B^{IV}C^{IV}$.

On the given grid plot a triangle ABC and its images $A^1B^1C^1$, $A^{11}B^{11}C^{11}$, $A^{111}B^{111}C^{111}$ and $A^{IV}B^{IV}C^{IV}$. And give coordinates of $A^{IV}B^{IV}C^{IV}$. (10mks)

19. A Post OT stand vertically on level ground John moves from O, the foot of the flag post to point R, on the level ground. The points T, O and R form a right angled isosceles triangle whose perimeter is 56m. S is another point on the level ground 35m from O calculate:

a) The angle of elevation of T from S. (6mks)

b) The distance ST. (2mks)

c) Find the maximum possible distance between R and S. (2mks)

20. A salesman received a basic salary of sh. 50,000 a year together with a commission of 6 % on the value of goods sold and a car allowance of sh. 2.50 per km

a) Find the total amount he received in a year in which he sells goods worth sh. 625,000 and travels 10,000km (4mks)

b) The next year he travels 12,000km and receives a total of shs. 134,000

i) Calculate the value of goods sold. (4mks)

ii) Calculate the percentage increase in the value of the goods sold. (2mks)

21. Two airports A and B are such that B is 500km due east of A. Two plane P and Q take off from A and B respectively and at the same time.

Plane P flies at 360km/hr on a bearing of 030°

Plane Q flies at 240km/hr on a bearing of 315°

The two planes land after 90 minutes.

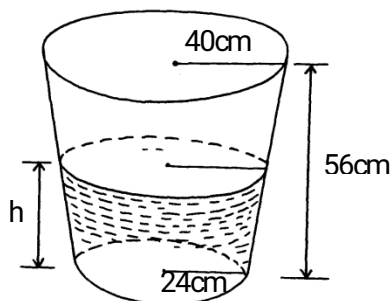
Using a scale of 1: 10,000,000

a) Show the positions of the planes after 90 min. (6mks)

b) Find the distance between the planes after 90 min. (2mks)

c) Find the bearing of plane Q from plane P after 90 minutes (2mks)

22. The figure below shows a container in form a frustrum of an open top radius 40cm and base radius 24 cm the depth is 56 cm



a) Calculate the volume of the container in litres. (4mks)

b) Of the container is $\frac{3}{4}$ full of water by volume,
Calculate the radius of the meniscus. (6mks)

23. Use a ruler and compass only in this question.

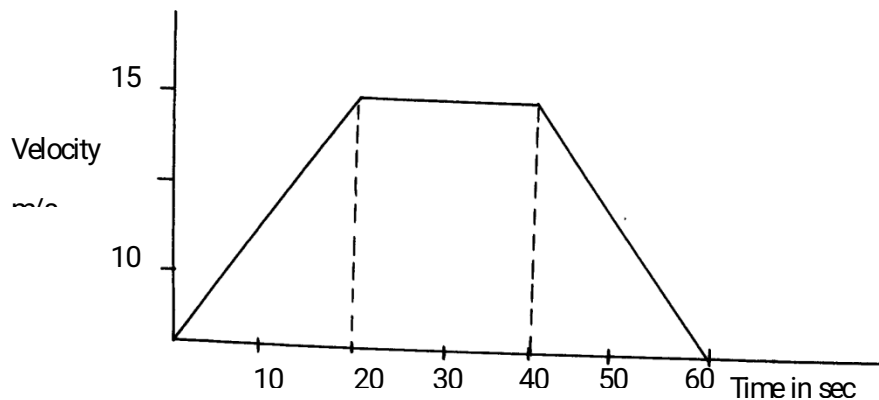
a) Construct $\triangle ABC$ such that $AB = 6\text{cm}$, $AC = 8.5\text{cm}$ and $\angle BAC = 120^\circ$ (3mks)

b) Construct the locus ℓ , of points equidistant from A and B (2mks)

c) Construct the locus ℓ of points equidistant from AB and BC (3mks)

d) Find the points of intersection, P_1 and P_2 , of ℓ_1 and ℓ_2 and measure P_1P_2 (2mks)

24. The diagram below shows the graph of a moving matatu from one bus stop to another.



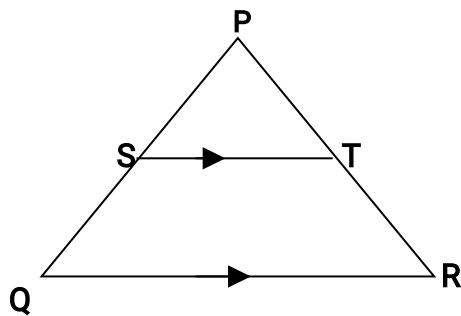
- Find the acceleration of the matatu. (2mks)
- Find the deceleration of the matatu (2mks)
- Calculate the distance the matatu while accelerating. (2mks)
- Calculate the distance the matatu covered while traveling at an acceleration of 0m/s^2 (2mks)
- Find the distance between the two bus stops. (2mks)

EXAM 3 PAPER 1

SECTION A (50 MARKS) Answer all questions in this section

- Solve the equation $\frac{x+1}{2} - \frac{x-3}{3} = 4$ (3mks)
- Find the equation of a line which passes through $(-1,-4)$ and is perpendicular to the line $y+2x-4=0$ (3mks)
- Given that $\cos A = \frac{5}{13}$ and A is acute angle. Find $\tan (90-A)$ (2mks)
- Simplify $\frac{12x^2-16x}{20-11x-3x^2}$ (4mks)

5. The figure below shows triangle PQR. PR=12cm TR=4cm and ST is parallel to QR



If the area of triangle PQR is 336cm^2 find the area of the quadrilateral QRTS. (4mks)

6. The internal radius of a pipe is 0.35m. Water flows through the pipe at a rate of 45cm per second. Calculate the amount of water that passes through the pipe in $2\frac{1}{4}$ hours in m^3 . (3mks)
7. A car moves with a constant acceleration of 8m/s^2 for 5 seconds. If the final velocity is 40m/s . calculate the initial velocity. (2mks)
8. Simplify without using a calculator (3mks)
- $$\frac{2\frac{1}{3} - 1\frac{2}{3} \div \frac{5}{9}}{\frac{4}{7} \text{ of } 2\frac{1}{3} - 2\frac{2}{7}}$$
9. Given that 4:5:8 is equivalent to 1:n:bn. Find the ratio of n:b only. (3mks)
10. Solve $32^{2x} \times 4^{x+3} = 128 \div 2^x$ (3mks)
11. John bought 3 books and 4 pens while Peter bought 4 books and 5 pens of the same type if Peter paid sh. 525 while John paid sh. 125 less than what Peter paid. Calculate the cost of each item (3mks)
12. A kenyan company received US dollars 100,000. The money was converted into Ksh. in a bank which buys and sells foreign currency as follows

	Buying Ksh	selling Ksh
1 US dollar	77.24	77.44
1 sterling pound	121.93	122.27

- (a) Calculate amount of money in Ksh. The company received (2mks)

(b) The Company exchanged the Ksh above into sterling pounds. Calculate how much it received to the nearest pound. (2mks)

13. (a) Use reciprocal tables to find the value of $= \frac{1}{0.325}$ (1mk)

(b) Hence, evaluate $3 \sqrt{\frac{0.000125}{0.325}}$ (2mks)

14. A point (2,3) is mapped onto (0,5) when rotated through angle of rotation 180° about centre (x,y). Find the coordinates of the centre. (3mks)

15. A square room is covered by a number of whole rectangular slabs of sides 60cm by 42cm. Calculate the least possible area of the room in square metres (3mks)

16. Using a ruler and pair of compasses only

(a) Construct a parallelogram PQRS in which PQ = 6cm QR = 4cm and $\angle SPQ = 75^\circ$ (3mks)

(b) Determine the perpendicular distance between PQ and SR (1mk)

SECTION II (50 MARKS)

Answer any five questions in this section in the spaces provided

17. A solid consists of a cone and a hemisphere. The common diameter of the cone and hemisphere is 12cm and the slanting height of the cone is 10cm

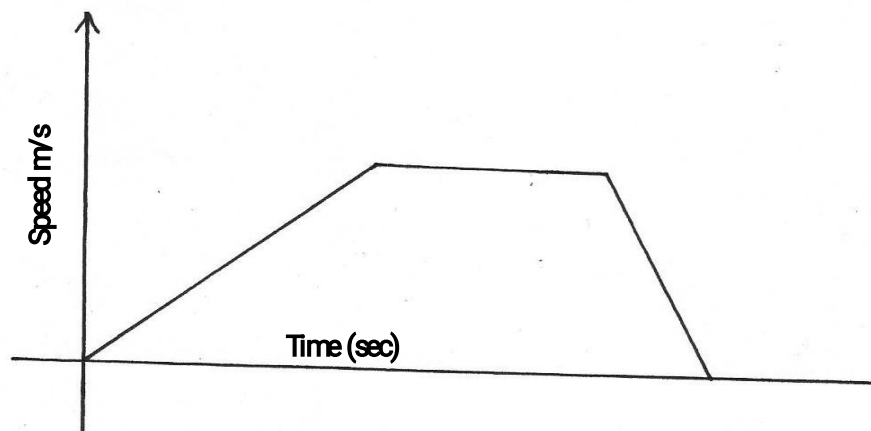
(a) Calculate the

(i) surface of the solid to 2 d.p (3mks)

(ii) the volume of the solid to 2 d.p (4mks)

(b) If the density of the material used to make the solid is 1.3g/cm^3 , calculate its mass in kilograms.

18. A cyclist moves from O to P covering a total distance of 3.75km. The cyclist starts from rest, accelerates uniformly for 240sec before moving at a constant speed for 100 sec and finally decelerates uniformly for 60 sec



(a) Determine the maximum speed attained by the cyclist. (3mks)

(b) Calculate the deceleration of the cyclist. (2mks)

(c) Calculate distance in metres, the cyclist travelled over the last 40 seconds (2mks)

(d) Determine the time in seconds, the cyclist took to travel $\frac{3}{5}$ of the journey (3mks)

19. Three trees R S and T are the vertices of a triangular field. R is 300m from S on a bearing of 300° and T is 450m directly south of R

(a) Using scale of 1cm rep 60m draw a diagram to show the positions of the trees. (3mks)

(b) Use the scale to determine

(i) the distance between T and S in metres (2mks)

(ii) the bearing of T from S (1mk)

(c) Find the area of the field, in hectares to 1 d.p

(4mks)

20. Two alloys A and B are each made up of copper, Zinc and tin. In alloy A, the ratio of copper to zinc is 3:2 and the ratio of zinc to tin 3:5

(a) Determine the ratio Copper: Zinc: tin in alloy A

(2mks)

(b) The mass of alloy is 250kg. alloy B has the same mass as alloy A but the amount of copper is 30% less than that of alloy A

Calculate

(i) the mass of tin in alloy A

(2mks)

(ii) The total mass of zinc and tin in alloy B

(3mks)

(c) Given that the ratio of zinc to tin in alloy B is 3:8 determine the amount of tin in alloy B than in alloy A
(3mks)

21. The table below shows the height of mango trees

Height	20-24	25-29	30-34	35-39	40-44	45-49	50-54	55-59
Freq.	2	15	18	25	30	6	3	2

(a) State the modal class

(1mk)

(b) Calculate to 2 d.p

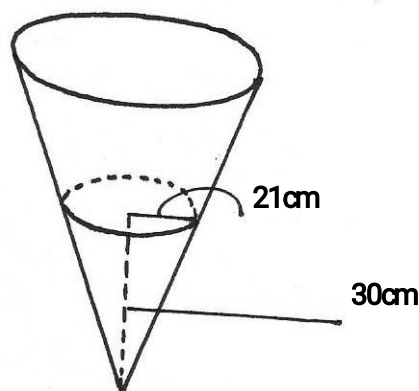
(i) The mean height

(4mks)

(ii) The difference between the median height and the mean height

(5mks)

22. The diagram below represents a conical vessel containing water to a depth of 30cm. the radius of the water surface is 21cm ($\pi = \frac{22}{7}$)



(a) Calculate the volume of the water in the vessel in cm^3

(2mks)

(b) Metal sphere is completely submerged in the water the level of water rises by 6cm

Calculate

(i) Radius of the new surface in the vessel

(2mks)

(ii) The volume of the metal sphere in cm^3

(3mks)

(iii) The radius of the sphere

(3mks)

23. (a) The ratio of Juma's and Akinyi's earnings was 5:3 Juma's earnings rose to sh.8400 after an increase of 12%.

(a) Calculate the percentage increase in Akinyi's earnings given that the sum of their new earnings is shs. 14,100.

(6mks)

(b) Juma and Akinyi contributed all the new earnings to buy maize at 1175 per bag. The maize was then sold at shs.1762.50. The two shared the money from the sales of the maize in ratio of their contributions. Calculate the amount that Akinyi got.

(4mks)

24. A triangle ABC has vertices A(3,4) B(1,3) and C(2,1)

(a) On the grid given draw triangle ABC

(1mk)

(b) Draw triangle A'B'C' the image of ΔABC under rotation of $+90^\circ$ about (0,0) and give its coordinates

(c) Draw triangle A''B''C'' the image of $\Delta A'B'C'$ under reflection in the line $y=x$ and give its coordinates

(d) By construction, find the triangle A'''B'''C''' the image of $\Delta A''B''C''$ after enlargement scale factor -2 centre (0,-1)

Give coordinates of the image A'''B'''C''' (4mks)

PAPER 2

1. Use tables of logarithms to evaluate $\left[\frac{6.195 \times 11.82}{83.52} \right]^{1/4}$ (3mks)

2. Solve the quadratic equation below $= \frac{4}{25}x^2 + \frac{4}{5}x + 1 = 0$ (3mks)

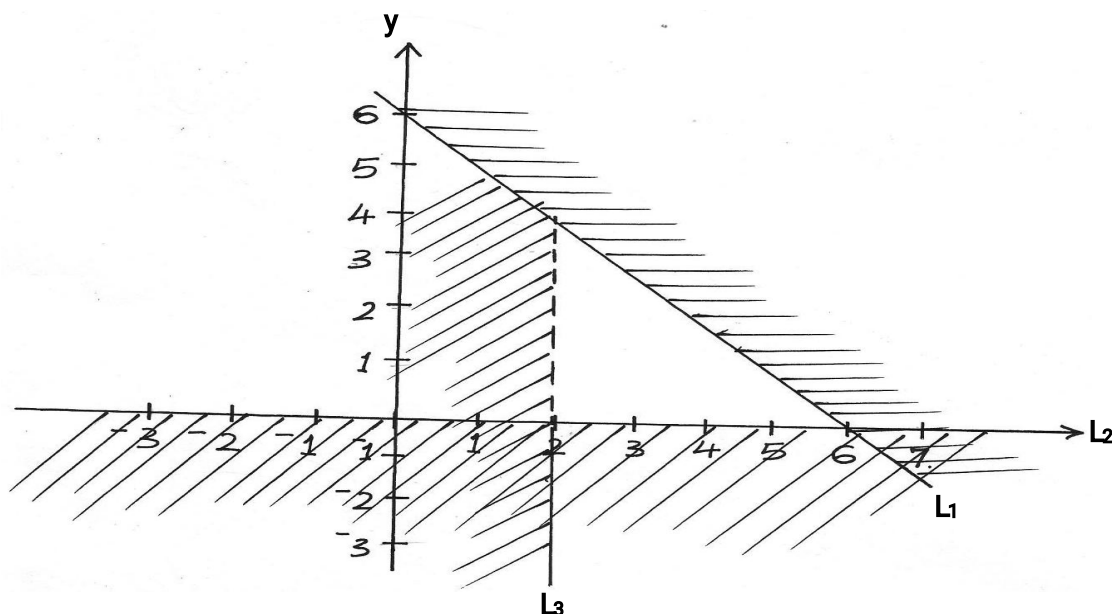
3. A box measures 6.4cm by 4cm by 5cm. find the limits within which its volume lies (3mks)

4. Ingwe and Okusimba live 8km apart. They left their homes at the same time and cycled at 8km/h and 7km/h respectively towards each other. Calculate the time they took before meeting. (2mks)

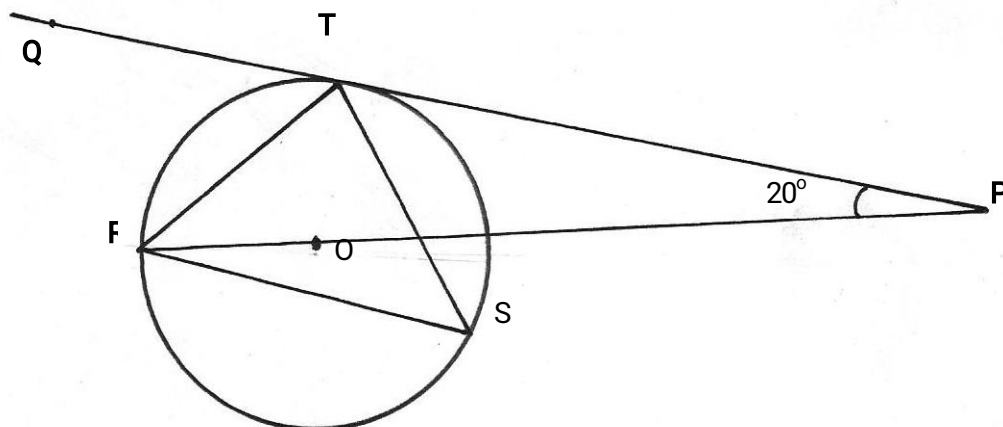
5. Given that $\sqrt{2} = 1.414$ and $\sqrt{3} = 1.732$ evaluate $\frac{5}{2\sqrt{2}-\sqrt{3}}$ (3mks)

6. Solve for x the equation $\log_2(x+3) = \log_2(x-2)$ (2mks)

7. Give the inequalities that define the unshaded area



8. In the figure below R, T and S are points on a circle centre O. PQ is a tangent to the circle at T, POR is a straight line and $\angle QPR = 20^\circ$. Find the size of $\angle RST$ (3mks)



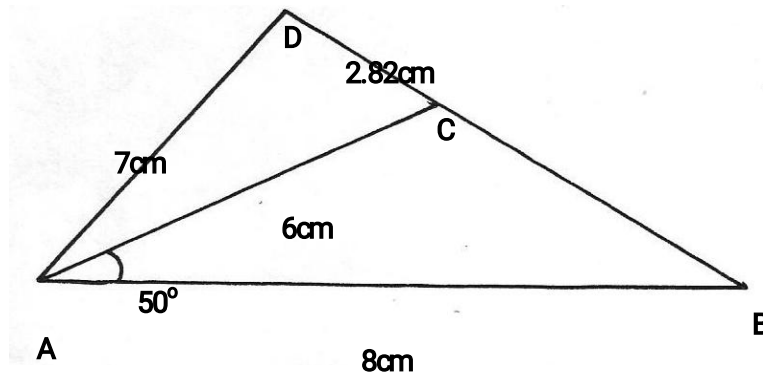
9. The size of interior angle of a regular polygon is $3x^\circ$ while its exterior angle is $(x-20)$. Find the number of sides of the polygon. (3mks)
10. Make h the subject of the formula (3mks)
- $$q = \frac{1+rh}{1-hr}$$
11. The third and the fifth term of an arithmetic progression are 10 and -10 respectively (2mks)
- (a) Determine the first and the common difference

- (b) The sum of the first 15 terms (2mks)
12. A watch which loses a half -minute every hour was set to read the correct time at 0545h on Monday. Determine the time in the 12 hour system, the watch will show on the following Friday at 1945h. (3mks)
13. (a) Expand $(1+x)^7$ up to the 4th term (2mks)
 (b) Use the expression in part (a) above to find the approximate value of $(0.94)^7$ (2mks)
14. Given that X varies directly as the square of y and $x=2$ when $y=1$, find x when $y=4$ (2mks)
15. A tea dealer mixes two brands of tea, x and y to obtain 35kg of the mixture worth Ksh.62 per kg. if brand x is valued at Ksh.68 per kg and brand y at Ksh 53 per kg . Calculate the ratio in its simplest form they are mixed. (3mks)
16. Obtain the centre and radius of the circle represented by the equation $x^2 + y^2 - 10y + 16 = 0$

SECTION II (50 MARKS)

Answer any five questions in this section in the spaces provided

17. In the figure below (not drawn to scale), $AB=8\text{cm}$, $AC=6\text{cm}$, $AD=7\text{cm}$, $CD=2.82\text{cm}$ and angle $CAB=80^\circ$



Calculate to 2 decimal places

- (a) the length BC (2mks)
- (b) the size of angle ABC (3mks)
- (c) the size of angle CAD (3mks)
- (d) the area of triangle ACD (2mks)
18. At the beginning of the year 1998, Okubasu bought two houses, one in Kakamega and other one in Mumias, each at Ksh.1,240,000 value of the house in Kakamega appreciated the rate of 12%
- (a) Calculate the value of the house in Kakamega after 9 years, to the nearest shillings (2mks)
- (b) After n years, the value of the house in Kakamega was Ksh. 2741245 while the value of the house in

Mumias was Ksh. 2917231

(i) Find n (4mks)

(ii) Find the annual rate of appreciation of the house in Mumias (4mks)

19. (a) Draw the graph for the function $y = -x^2 + 4x - 1$ for $-2 \leq x \leq 6$ (5mks)

(b) On the same axes draw the graph of $y = 2x - 3$ (1mk)

(c) Use your graph to solve the following equation

(i) $x^2 - 4x + 1 = 0$ (2mks)

(ii) $x^2 - 2x - 2 = 0$ (2mks)

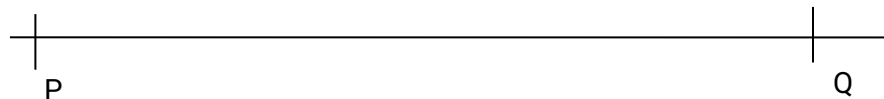
20. Using a ruler and pair of compasses only, construct a parallelogram PQRS in which $PQ = 8\text{cm}$, $PS = 5\text{cm}$ and angle $SPQ = 45^\circ$

(a) Measure SQ

(b) (i) Construct a perpendicular bisector of the line S and Q

(ii) Construct a perpendicular bisector of line QR to meet the one constructed in b(i) above at point O.

(iii) Draw a circle passing through points S, Q and R and measure its radius



21. (a) Find A^{-1} , the inverse of matrix $A = \begin{pmatrix} 5 & 6 \\ 7 & 9 \end{pmatrix}$ (2mks)

(b) Asira bought 5 books and 6 sets for a total of Ksh. 2440. Awino bought 7 books and 9 sets for a total

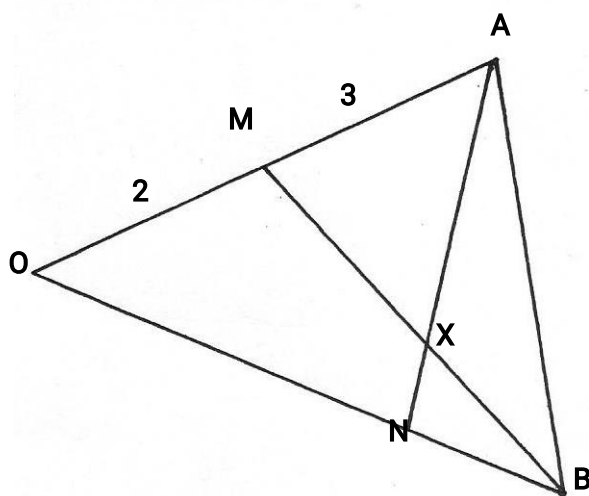
of Ksh. 3560.

(i) Form a matrix equation to represent the information (1mk)

(ii) Use matrix method to find the price of a book and that of a set. (3mks)

(c) A school bought 36 books and 50 sets. A discount of 5% was allowed on each book whereas 8% on each set. Calculate the percentage discount on the cost of all the items bought by the school (4mks)

22. The figure below shows triangle OAB in which M divides OA in the ratio 2:3 and N divides OB in the ratio 4:1



(a) Given that $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$, express in terms of $\mathbf{a}$ and $\mathbf{b}$

(i) $\vec{AN}$ (1mk)

(ii) $\vec{BM}$ (1mk)

(b) Express $\vec{OX}$ in terms of $\mathbf{a}$, $\mathbf{b}$, t and s given that $\vec{AX} = s\vec{AN}$ and $\vec{BX} = t\vec{BM}$ and t and s are constants (2mks)

(c) Find the value of s and t hence write $\vec{OX}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ (6mks)

23. The probability that Hillary, Leonard and Charles will be late for breakfast on any one morning are $\frac{1}{4}$, $\frac{1}{3}$ and $\frac{1}{5}$ respectively on any one morning

(a) Draw a probability tree diagram to represent the information. (2mks)

(b)(i) none of them will be late (2mks)

(ii) Only one of them will be late (2mks)

(c)(i) at least one of them will be late (2mks)

(ii) at most one of them will be late

(2mks)

24. A hot water tap can fill a bath in 5 minutes while a cold water tap can fill the same bath in 3 minutes. The drain pipe can empty the full bath in $3\frac{3}{4}$ minutes. The two taps and the drain pipe are fully open for $1\frac{1}{2}$ min after which the drain pipe is closed. How much longer will it take to fill the bath?
(10mks)

EXAM 4

PAPER 1

SECTION I (50 MARKS)

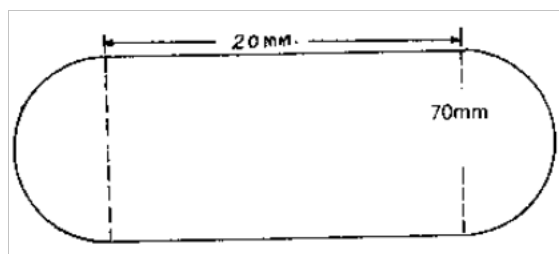
Attempt ALL questions from this section

1. Show that 8260439 is exactly divisible by 11, using test of divisibility. (2mks)
2. Use logarithm tables to evaluate

$$\sqrt[3]{(4.562 \times 0.038)(0.3+0.52)^{-1}}$$

Giving your answer to 3 significant figures. (4mks)

3.



The above figure (not drawn to scale) shows the cross-section of a metal bar of length 3 meters. The ends are equal semi-circles. Determine the mass of the metal bar in kilograms if the density of the metal is 8.87g/cm^3 .

(3mks)

4. A surveyor recorded the information about a tea farm in his field book as in the table below.

	Q	
	600	90 to C
To A 180	420	
	300	90 to D
To B 50	50	
	P	

- (a) Given that $PQ = 65\text{m}$, make a sketch of the field. (2mks)

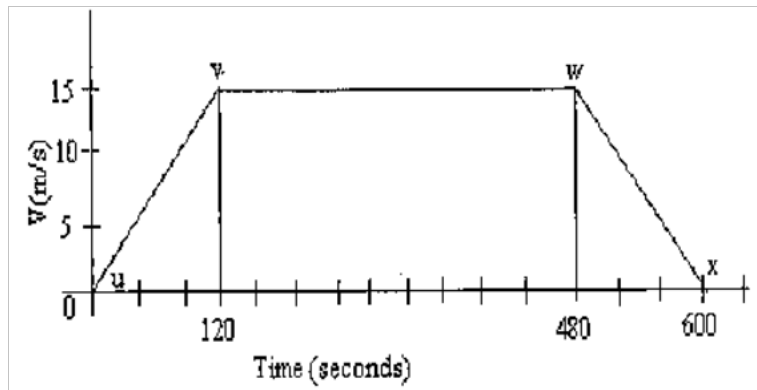
Hence find the area of the field in hectares.

(2mks)

5. Tap A fills a water tank in 30min, B in 20mins and C in 10mins. All three taps are turned on from 8:55am to 8:59am and then C is turned off. At what time will the tank be filled after C has been closed? (3mks)

6. The mean of a set of n numbers is 28. If an extra number 18 is included in the set, the mean now becomes 26. Find the value of n . (2mks)

7. The graph below shows the motion of a bus for 10 minutes



The bus moves from V to X. Calculate the distance (total distance) moved by the bus from V to X and hence its average speed. (3mks)

8. Solve the equation $\frac{x+1}{2} = 1 - \frac{1-3x}{5}$ (2mks)

9. A tailor brought two pair of trousers at sh. 1600. He marked the price such that after allowing his discount of 15%, he would still make a profit of 30% on the cost price. Determine the price at which a pair of trouser was marked. (3mks)

10. A two digit number is made by combining any 2 of the digits 1, 3, 5, 7, 9, at random

(a) Make an array of possible combinations (2mks)

(b) Find the probability that the number is a prime number. (2mks)

11. Find the value of X in the following equation.

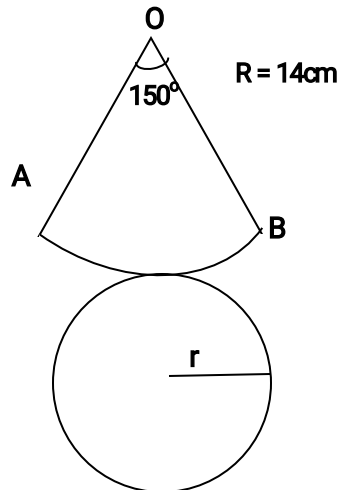
$$25^{x-1} + 5^{2x} = 130 \quad (4mks)$$

12. Solve the simultaneous equations (3mks)

$$\text{Log}_4(2x + y) = 2$$

$$\text{Log}_3(5x + 3y) = 2$$

13. Given $A(3,6)$ $B(9,10)$ and that a point R divides AB in the ratio $3 : -2$, find the co-ordinates of R .
14. The figure below shows a net of a circular cone with a lid. Given that $\angle AOB = 150^\circ$ and $OA = 14\text{cm}$.



Determine

- (a) The radius of the base of the cone. (2mks)
- (b) The total, surface area of the cone $\left(\pi = \frac{22}{7} \right)$ (2mks)
15. 28% of Charles salary goes to taxes. The remainder is Charles' disposable income. He uses $\frac{1}{4}$ of it for food and $\frac{3}{4}$ of the remainder for travelling and saves the rest. If the difference between what Charles saves and what goes for taxes is sh. 9860. How much does Charles earn before taxation.
16. Find the integral values of x for which $5 \leq 3x + 2$ and $3x - 14 < -2$ (3mks)

SECTION II (50 MARKS)

Answer only five questions in this section in the spaces provided.

17. Mr. Omwega is employed. His basic salary is Kshs. 21, 750 and is entitled to a house allowance of Kshs 15, 000 and travelling allowance of Kshs. 8, 000 per month. He also claims a family monthly relief of Kshs. 1, 056 per month. Other deductions are;
- Union dues Kshs. 200 and
- Co-operative shares Kshs 4, 500 per month.

The table below shows the tax rates for the year.

Income (Kshs per annum)	Tax rates
-------------------------	-----------

1 – 116, 600	10%
116, 161 – 225, 600	15%
225, 601 – 335, 040	20%
335, 041 – 444, 480	25%
Over 444, 480	30%

Calculate;

- (a) Mr. Omwanga's annual taxable income. (2mks)
- (b) The tax paid by Mr. Omwanga in the year. (6mks)
- (c) Mr. Omwanga's net income per month. (2mks)

18. Using a ruler and a pair of compasses only. Construct a parallelogram **ABCD** such that **AB** = 8cm diagonal **AC** = 12cm and angle **BAC** = 22.5° (4mks)

- (a) Measure
 - (i) The diagonal **BD** (1mk)
 - (ii) The angle **ABC** (1mk)
- (b) Draw the circumcircle of triangle **ABC** (2mks)
- (c) Calculate the area of the circle drawn. (2mks)

19. In the figure below, **O** is the centre of the circle. **PQR** is the tangent to the circle at **Q**. Angle **PQS** = 28° , angle **UTQ** = 54° and **UT** is a tangent to the circle at **T**. Find the size of angle **TSQ**.

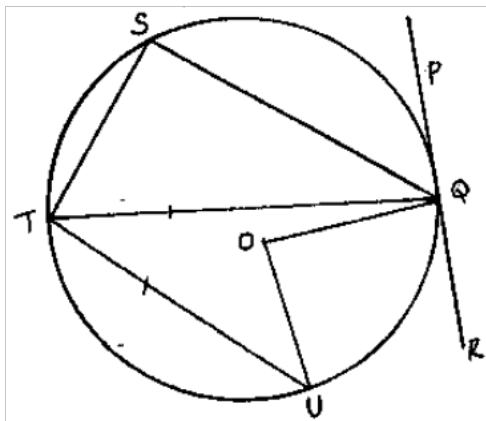


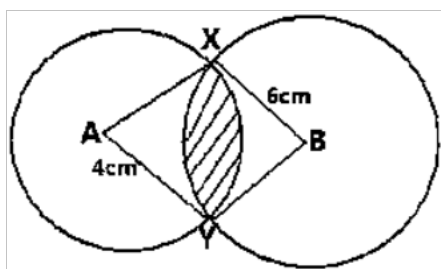
figure below, **O** is the centre of the circle. **PQR** is the tangent to the circle at **Q**, Angle **PQS** = 28° , angle **UTQ** = 54° and **UT** is a tangent to the circle at **T**. Find the size of angle **TSQ**.

Giving reasons, determine the size of

- (a) Angle **STQ** (2mks)
- (b) Angle **TQU** (2mks)
- (c) Reflex angle **TQS** (2mks)
- (d) Reflex angle **UOQ** (2mks)
- (e) Angle **TQR** (2mks)

20. Two circles of radii 4cm and 6cm intersect as shown below. If angle **XBY** = 30° and angle **XAY** = 97.2°.

Find the area of the shaded part. (Take $\pi = \frac{22}{7}$) (10mks)



21. A bag contains 5 red, 4 white and 3 blues beads. Two beads are selected at random.

- (a) Draw a tree diagram and list the probability space. (3mks)
- (b) Find the probability that
 - (i) The last bead selected is red. (2mks)
 - (ii) The beads selected were of the same colour. (2mks)
 - (iii) At least one of the beads is blue. (3mks)

22. (a) Complete the table below for the function $y = x^2 + \frac{12}{x} - 15$ for $0.5 \leq x \leq 4$ (2mks)

X	0.5	1	1.5	2	2.5	3	3.5	4
Y	9.25			-5	-4			

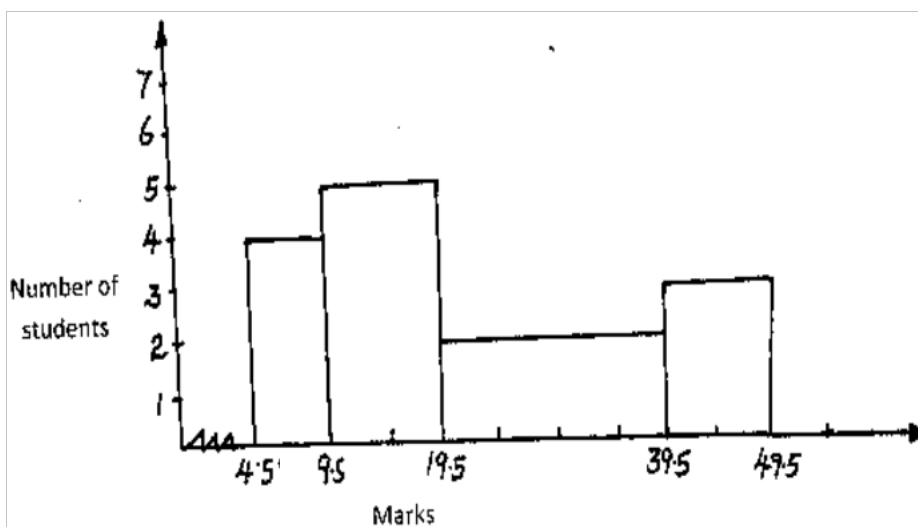
(b) Draw the graph of $y = x^2 + \frac{12}{x} - 15$ for $0.5 \leq x \leq 4$, using a scale of 2cm = 1 unit on the x-axis and 2cm for 5 units on the y-axis (3mks)

(c) (i) From your graph, state the range of values of x for which $y = x^2 + \frac{12}{x} \leq 18$ (2mks)

(ii) By adding a suitable straight line to your graph, solve the equation

$$y = x^2 + \frac{12}{x} - 5x + 20$$

23. The diagram below shows a histogram representing the marks obtained in a certain test.



(a) Prepare a frequency distribution table for the data. (4mks)

(b) Estimate

(i) the mean mark (4mks)

(ii) the median mark (2mks)

24. triangle OAB is such that $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$. point A divides OP in the ratio 5 : -2 and point Q is on AB such that $\vec{AQ} = \frac{1}{5} \vec{AB}$. OQ and BP intersect at T .

(a) Draw a diagram to represent the given situation. (1mk)

(b) Express in terms of $\mathbf{a}$ and $\mathbf{b}$

(i) $\vec{OQ}$ (1mk)

(ii) $\frac{BP}{OQ}$ (1mk)

(c) Given that $OT = kOQ$ and $BT = hBP$ where k and h are constants.

(i) Express $\frac{OT}{OQ}$ in two different ways.

~

(ii) Determine the values of k and h . (3mks)

(iii) Determine the ratio in which T divides OQ (1mk)

PAPER 2

SECTION I (50MARKS)

Answer ALL questions from this section

1. Find the **GCD** of 60, 100 and 840 (2mks)

2. Evaluate $1\frac{3}{4} \times 4\frac{1}{2} - \frac{1}{2} \div 2\frac{3}{5} + \frac{2}{5}$ (3mks)

3. Find the 10th term and the sum of the first ten terms of the GP (4mks)

4. Five people can build 3 huts in 21 days. Find the number of people working at the same rate that can build 6 similar huts in 15 days. (2mks)

5. The distance covered by a rolling object at an interval of 1 second were recorded as 4cm, 16cm, 28cm, 40cm etc. After how long will the object be 22.6M from the starting point? (4mks)

6. A bus service number 4 leaves a terminus every 15 minutes. Services 8 and 3 leaves every 20 and 30 minutes respectively. If all the three services leave together at 06.00 am, at what other times during day time do they leave together? (3mks)

7. The length and breadth of a rectangle are measured and given as 2.5cm and 1.2cm respectively. Determine the maximum possible error in the area calculated using these values. (4mks)

8. Find the eighth term and the sum of the first 15 terms of the **AP**. (4mks)

$$-7 + -4 + 1 + \dots$$

9. An arc of a circle, radius 10cm, subtends an angle of 80° at the centre of the circle. Find the length of the arc. (2mks)
10. In a trapezium of area 40cm^2 the ratio of the lengths of the parallel sides is 2 : 3. Find the lengths of the parallel sides if they are 5cm apart. (3mks)
11. Find the volume of a prism 20cm long and whose cross-section is an equilateral triangle of side 9cm.
12. The equations of a line L_1 is $4y - 3x + 8 = 0$. Find the equation of line L_2 which is perpendicular to L_1 and passes through (2, 7) (3mks)
13. Solve for x in $3^{2x+1} + 5(3^x) - 2 = 0$ (3mks)
14. Express $\frac{2 - \sqrt{3}}{2 + \sqrt{3}} + \frac{4 - \sqrt{3}}{2 - \sqrt{3}}$ in the form $a + b\sqrt{c}$ (3mks)
15. Expand $(1 + \frac{1}{2}x)^7$ up to the first four terms. Hence, use your expansion to find the value of $(1.02)^7$ (4mks)
16. Y varies directly as the square of x and inversely as z. If x is increased by 10% and z is decreased by 20%, find the percentage change in Y (4mks)

SECTION B

Answer only five questions in this section.

17. Given that y is inversely proportional to x^p and k is the constant of proportionality
- (a) (i) Write down a formula connecting y, x, n and k (1mk)
- (ii) If $x = 2$ when $y = 12$ and $x = 4$ when $y = 3$, write down two expressions for k in terms of n. Hence find the value of n and k (7mks)
- (b) Using the value of n obtained in (a) (ii) above find y when $x = 5\frac{1}{3}$ (2mks)
18. (a) After getting 10% discount, a man pays Ksh. 450 for a sweater. Find the marked price of the sweater.
- (b) After how long will Kshs. 1, 000 grow to Kshs. 2, 000 at 18% p.a. simple interest ?

- (c) Find the compound interest on Kshs. 10, 000 for 2 years at 12 % p.a.
- (d) A man invests Kshs. 10, 000 in an account which pays 16% interest p.a. The interest is compounded quarterly. Find the amount in the account after $1\frac{1}{2}$ years.
19. A water vendor has a tank of capacity 18900 liters. The tank is being filled with water from two pipes A and B which are closed immediately the tank is full. Water flows at the rate of $150000\text{cm}^3/\text{minute}$ through pipe A and $120000\text{cm}^3/\text{minute}$ through pipe B.
- (a) If the tank is empty and the two pipes are opened at the same time. Calculate the time it takes to fill the tank. (3mks)
- (b) On a certain day the vendor opened two pipes A and B to fill the empty tank. After 25minutes he opened the outlet pipe to supply water to the consumers at an average rate of 20 litres per minute.
- (i) Calculate the time it took to fill the tank on that day. (4mks)
- (ii) The vendor supplied a total of 542 jericans each containing 25 liters of water on that day. If the water that remained in the tank was 6300 liters calculate in litres the amount of water that was wasted. (3mks)

20. The frequency distribution table shows marks obtained by 40 pupils in a mathematics test.

Marks	30-39	40-49	50-59	60-69	70-79	80-89	90-99
Number of pupils	2	3	10	12	8	3	2

- a) State the modal class (1mk)
- b) calculate the mean mark (5mks)
- c) Calculate the median mark (4mks)
21. Kisumu boys High School hall has 200 seats. During the District Drama festivals, tickets were sold at Ksh. 150 for adults and kshs. 75 for students.
- (a) On day one of the festival 80% of the seats in the hall were occupied and twenty of the seats were occupied by students. Calculate the total money collected from the sale of tricketts this day. (3mks)
- (b) On the last day of the festival, x students occupied the seats and all seats were occupied. The money collected from the sales was sh. 25, 350.

(i) Write down an equation of x (2mks)

(ii) Calculate the value of x (2mks)

(c) The money collected from the sale of tickets during the festivals was divided among cost of hosting, allowances for adjudicators and electricity bill in the ratio of 7 : 3 : 2, If the allowances amounted to 126,000, calculate the:-

(i) Amount collected during the festival (3mks)

(ii) The cost of electricity bill during the festival (1mk)

22. Triangle PQR has vertices at $P(3, -1)$, $Q(5, 2)$ and $R(2, 3)$. Plot and draw PQR on the grid provided (1mk)

(b) Given that triangle $P'Q'R'$ is the image of PQR under positive quarter turn about the origin, and draw $P'Q'R'$ on the same axis as PQR (3mks) plot

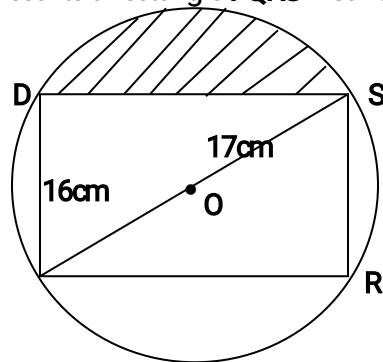
(c) $P''Q''R''$ is the image of PQR after a reflection in the line $y + x = 0$. Plot and draw $P''Q''R''$ on the same axes as PQR and $P'Q'R'$ above. (3mks)

(d) State the pairs of triangles above that are:

(i) Oppositely congruent (2mks)

(ii) Directly congruent (1mk)

23. The figure below represents a rectangle $PQRS$ inscribed in circle centre O and radius 17cm $PQ = 16$ cm



Calculate

(a) the length PS of the rectangle (3mks)

(b) the angle POS (3mks)

(c) the area of the shaded region (4mks)

24. Data collected from an experiment involving two variables x and y was recorded as shown in the table below.

X	1.1	1.2	1.3	1.4	1.5	1.6
Y	-0.3	0.5	1.4	2.5	3.8	5.2

The variables are known to satisfy a relation of the form $y = ax^3 + b$ where a and b are constants

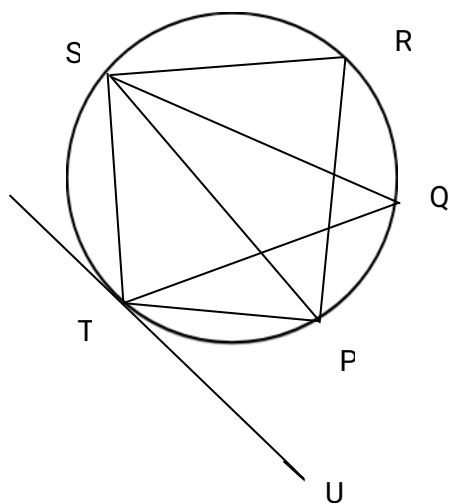
- (a) For each value of x in the table above write down the value of x^3 (2mks)
- (b) (i) By drawing a suitable straight line graph estimate the value of a and b . (7mks)
- (ii) Write down the relationship between y and x (1mk)

EXAM 5 PAPER 1

SECTION I (50 MARKS)

Answer ALL Questions in this Section

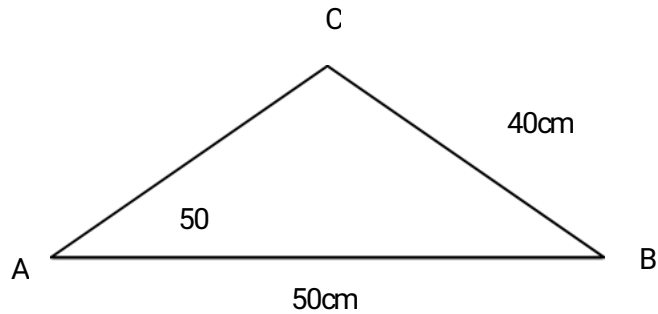
1. Given that $\log_{10}3 = 0.4771$ and $\log_{10}7 = 0.8451$, evaluate $\log_{10}0.63$ without using mathematical tables or calculator. (3marks)
2. Simplify completely the expression: $\frac{(2-x)(x+3)}{x^2+3x-10} + \frac{x+8}{x+5}$. (3marks)
3. In the diagram below TU is a tangent to the circle at T. Given that angle SRP = 130° and angle SQT = 70° , calculate the size of angle PTU. (3marks)



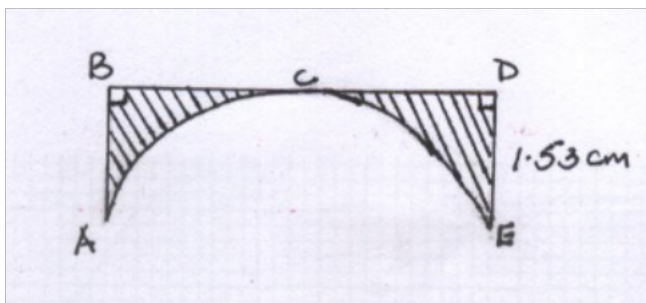
4. The figure below is a triangle in which angle $BAC = 50^\circ$, $AB = 50$ cm and $BC = 40$ cm. Calculate the area of the

triangle.

(3marks)



5. Find the median of the following data:
10, 2, 7, 5, 9, 6, 18, 4, 6, 3, 7, 8. (2marks)
6. Two perpendicular lines pass through the point (0, -4). If the equation of one of the lines is $2y = 3x - 8$, determine the equation of the other. (3marks)
7. A salesman earns 3% commission for selling a chair and 4% commission for selling a table. A chair fetches sh 600 and a table goes for sh1500. One time, he sold 10 more chairs than tables and earned sh 7200 as commission. Find the number of chairs and tables sold (4marks)
8. A super market dealer in Nairobi mixes 100kg of local brown sugar with 80kg of imported white sugar. If the dealer sold 270kg of mixed sugar at a total of sh. 15,480, calculate the percentage increase in the total sales due to selling mixed sugar if a 1kg packet of brown sugar sells at sh. 50 and a 1kg packet of white sugar sells at sh. 45. (3marks)
9. A regular polygon has internal angle of 150° and a side of length 10cm.
- (a) Find the number of sides of the polygon (2marks)
- (b) Find the perimeter of the polygon (1mark)
10. The figure below shows an arc ACE of a circle with centre O and radius 6cm. If $BC=CD=4\text{cm}$, calculate the area of the shaded region if $DE=AB=1.53\text{cm}$ (4marks)



11. Evaluate without using mathematical table or calculator $\frac{0.021 \times 0.246 \times 1.75}{11.48 \times 0.014}$ expressing the answer as a fraction in its simplest form. (2 marks)
12. An American tourist arrived in Kenya with 1000 US \$ and converted the whole amount into Kenyan shillings. He spent sh. 40,000 and changed the balance to sterling pound before leaving for United Kingdom. A Kenya bank buys and sells foreign currencies as shown.

	Buying (in Kshs)	Selling (in Kshs)
1 US dollar	84.2083	84.3806
1 Sterling Pound	134.7941	135.1294

Calculate the amount he received to the nearest sterling pound. (3 marks)

13. Solve the following inequality and represent your solution on a number line. $3 + x - 4(x - 3) \leq 30$ (3 marks)
14. A rectangular petrol tank A, is 50cm high and holds 16 litres of petrol when it is $\frac{2}{5}$ full. A similar tank, B is 80cm high. Calculate
- Base area of tank A. (2 marks)
 - Capacity of tank B. (2 marks)
15. By using the substitution $y = 2^x$, solve the equation.
 $5 \times 2^{2x+1} - 3 \times 2^x - 34 = 0$ (3 marks)
16. Pipe A empties a full tank in 8 hours, pipe B empties in 6 hours and pipe C empties in 3 hours. If all the three pipes are open for 30 minutes and then pipe A closed, How long will it take to empty a full tank. (3 marks)

Section II (50 Marks)

Answer any **five** questions in this section

17. Two farmers Ndume and Njawa took a total of 40 beef cattle for sale at two different slaughter houses and each got the same amount from the sale. If Ndume had sold at Njawa's selling price, he would have got sh. 115,200 and if Njawa had sold at Ndume's selling price, he would have got sh. 51,200.

a) How many beef cattle did Ndume take to the slaughter house? **(8marks)**

b) What was the buying price at the slaughter house where Njawa took his cattle?
(2marks)

- 18.a) Find the equation of a straight line passing through the points (3,2) and (-3,6) giving your answer in the form

$$\frac{x}{a} + \frac{y}{b} = 1, \text{ where } a \text{ and } b \text{ are constants.} \quad \textbf{(3marks)}$$

(b) State the coordinates of point A and B, at which the line in (a) above crosses the x-axis and y-axis respectively.
(2marks)

(c) Find the equation of the median from O, the origin, to AB. **(3marks)**

(d) Find the angle the line in (a) above makes with the positive x-axis. **(2marks)**

19. In a physics test student scored the following marks.

72 50 43 58 62 49 69 60 84 62 55

89 67 92 81 75 63 77 95 65 54 35

45 73 41 56 50 36 49 58 61 85

38 64 76 78 51 43 72 37 62 55

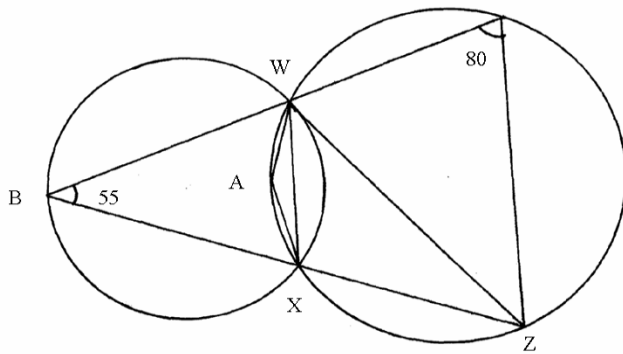
(a) Using a class width of 10 and 35-44 as the first class, make a frequency table of the grouped data.
(5marks)

(b) Estimate

(i) the mean **(2marks)**

(ii) the median **(3marks)**

20. The figure below shows two circles that intersect at point W and X. Point A is the centre of smaller circle and lies on the circumference of the larger circle WXZY and BWY are straight lines. Angle WBX = 55° and angle WYZ = 80° .

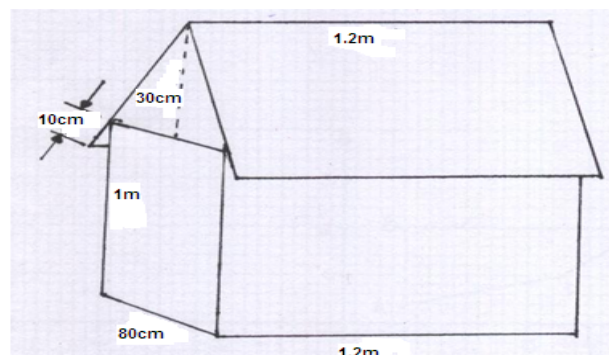


Find the following angles giving reasons.

- (a) The obtuse angle WAX. (2 marks)
- (b) Angle WYZ (2 marks)
- (c) Angle XWZ (2 marks)
- (d) Angle WXA (2marks)
- (e) Angle AXB (2 marks)

- 21.a) The second, fourth and eighth terms of an arithmetic progression are in geometric progression, and the sum of the third and fifth term of the arithmetic progression is 20. Find the sum of the first six terms of the arithmetic progression. (5marks)
- (b) A commercial plot is valued at sh. 500,000. The plot depreciates at a rate of 10% per six months for a period of 2 years. It then appreciates at a rate of 4% per quarter yearly for 3 years. Find the amount by which the value of the plot changed after 5 years. (5marks)

22. The figure below shows a portable Kennel.



- a) Calculate
- i) The total surface area of the walls and the floor. (2marks)

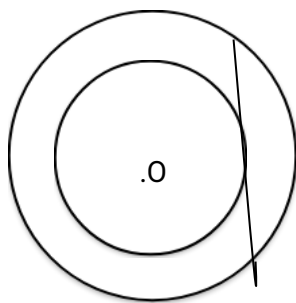
- ii) The total area of the roof. (2marks)
- b) The cost of roofing is Ksh 500 per square meter and that of making walls and floor is Ksh 600 per m^2 . Find the cost of making the Kennel. (2marks)
- c) Find the cost of roofing another Kennel whose dimensions are 50% more than those of the given Kennel. (2marks)
- d) Calculate the volume of air inside the Kennel in cm^3 . (2marks)
23. Every Sunday Alex drives a distance of 80 km on a bearing of 074° to pick up his brother John to go to church. The church is 75 km from John's house on a bearing of $S 50^\circ E$. After church they drive a distance of 100 km on a bearing of 260° to check on their father before Alex drives to John's home to drop him off then proceeds to his house.
- (a) Using a scale of 1 cm to represent 10 km, show the relative positions of these places. (4 marks)
- (b) Use your diagram to determine:
- (i) the true bearing of Alex's home from their father's house. (1 mark)
- (ii) the compass bearing of the father's home from John's home. (1 mark)
- (iii) the distance between John's home and the father's home. (2 marks)
- (iv) the total distance Alex travels every Sunday. (2 marks)
24. The vertices of triangle PQR are P (0, 0), Q (6 0) and R (2, 4)
- a) Plot these points on the grid provided below. (1 mark)
- b) Triangle $P^1Q^1R^1$ is the image of a triangle PQR under an enlargement scale factor $\frac{1}{2}$ and centre (2, 2). Write down the coordinates of triangle $P^1Q^1R^1$ and plot on the same grid. (2 marks)
- c) Draw triangle $P^{11}Q^{11}R^{11}$ the image of triangle $P^1Q^1R^1$ under a positive quarter turn about the point (1, 1).
- d) Draw a triangle $P^{111}Q^{111}R^{111}$ the image of triangle $P^1Q^1R^1$ under reflection in the line $y = 1$. (3 marks)
- e) Describe fully a single transformation which maps triangle $P^{111}Q^{111}R^{111}$ onto triangle $P^1Q^1R^1$. (2 marks)

PAPER 2

SECTION I (50 MARKS)

Answer ALL Questions in this Section

1. If the values of x and y in the ratio $(2x^2 + 3y^2) : (2x^2 - 3y^2) = 5 : 3$ also satisfy a quantity M given by $M = \frac{10x^2 + 21y^2}{9y^2 + 3x^2}$. Find the value of M . (3 marks)
2. The distance, y units, travelled by an ant as it traces the path ought to have been followed by the queen inside an ant-hill, measured from the ground level (x -axis) is given by $y = 5x^2 - 30x + 47$. By expressing y in the form $y = k(x + m)^2 + n$ where k , m and n are integers, state the values of k , m and n . (3marks)
3. A man deposits sh. 500,000 in an investment which pays 12% per annum interest compounded quarterly per year. Find the accumulated interest in his account after 4 years. (3 marks)
4. In an experiment to determine a local value of acceleration of free fall g , a student made 11% error and 2.5% error in measuring length l and computing periodic time T respectively of a pendulum. He also made a 2% error in approximating the value of π . If $g = \frac{4\pi^2 l}{T^2}$, calculate the percentage error in obtaining g . (2marks)
5. In a certain competition sweets are placed 6m apart from a point A up to another point B. There are 30 sweets in total. A container is placed at a point C, 30m before A. The competitor picks one sweet at a time and takes it to the container C, then goes for another sweet, until all are finished. If he starts at point C, determine the total distance travelled by the competitor. (3marks)
6. The co-ordinates of points A and B are $(-3, 6)$ and $(5, 24)$ respectively. If C is a point on the line AB such that $AC : CB = 9 : 5$, find the coordinates of C. (3 marks)
7. If $p = 5 + \sqrt{2}$ and $q = 3 - 4\sqrt{2}$ find the value of $\frac{q}{p + q}$ in the form $a + b\sqrt{c}$ where a , b and c are rational numbers. (3marks)
8. The air resistance, R , against a body in motion varies partly directly to the square of velocity, v , and partly inversely as the cross-sectional area, A , of the object. When the velocity is 20 m/s and area is $2m^2$, the air resistance is 120,000 N and when the velocity is 50 m/s and area is $4m^2$, the air resistance is 520,000 N. Calculate the air resistance when the velocity is 40 m/s and area is 2.5 m^2 . (4 marks)
9. Solve by completing the square: $6x^2 + 11x - 10 = 0$. (3 marks)
10. Two concentric circles have a radius of 6cm and 7.2 cm respectively. Find the length of the chord of the outer circle which is a tangent to the inner one. (3marks)



11. A cone of radius 3 cm is made from a sector of a circle. If the area of the sector is 66 cm^2 , calculate the volume of the cone. (4marks)

12. A point P (4, 6) is mapped onto a point $P^1 (8 - 4)$ under a translation. If $Q^1 (5, 3)$ is the image of a point Q under the same translation, find the co-ordinate of Q. (3marks)

13. Solve the trigonometric equation below for $0 \leq \theta \leq 360^\circ$.

$$4 \sin \theta + 5 \cos \theta = 0.$$

(3marks)

14. Make x the subject of the formula: $T = \frac{Ex}{x^2 + n}$.

(3marks)

15. A triangle ABC is such that AB = 10 cm and BC = 12 cm and angle ABC = 100° . Calculate the diameter of its circumcircle. (3marks)

16. Solve the equation $\log (x + 2) = \log 4 - \log (x + 5)$. (3marks)

Section II (50marks)

Answer any five questions in this section

17.a) Using a matrix method solve the following equation. (4 marks)

$$3x + 2y = 6$$

$$4y + 10x = 4$$

b) If $P = \begin{pmatrix} 6 & 3 \\ 1 & 1 \end{pmatrix}$ and $Q = \begin{pmatrix} \frac{1}{3} & -1 \\ -\frac{1}{3} & 2 \end{pmatrix}$

i) find PQ . (2 marks)

ii) Hence find a^{-1} . (1 mark)

c) A shopkeeper buys 2 bags of sugar, 4 bales of flour and 3 bales of salt. He already has 1 bag of sugar, 2 bales of flour and 2 bales of salt. He sales each of bag of sugar at Ksh. 3,200, each bale of flour at Ksh. 600 and each bale of salt at Ksh. 500. Using matrix addition, subtraction and multiplication, determine his income after selling all the mentioned items. (3 marks)

18. The table below shows Kenya's tax rates a certain year.

Income (K£ p.a)	Tax rates (Ksh Per £)
1-5220	2
5221 – 10440	3
10441 – 15660	4
15661 – 20880	5
20881 and above	6

In that, Mr. Omulloh lived in a company house for which he paid a nominal rent of sh. 1250 per month. He enjoyed an automatic personal relief of sh. 1,056 per month and a further life insurance relief of k. 600 per month. It is also known that he paid a per month net tax of sh. 3930. If his salary per year was sh. Y and his total taxable income was sh. Z per year,

- a) Express Z in terms of Y. (2 marks)
- b) How much tax was charged on sh. Z before any relief was given that year? (2 marks)
- c) Use the table of tax rates above to determine the value of Z (4 marks)
- d) Find the value of Y. (2 marks)

19.a) Musa obtained a loan in which the rate of interest was charged at 3.5% quarterly. He cleared this loan by paying Kshs 24,805 at the end of $1\frac{1}{2}$ years.

How much was the loan if the rate was on:

- i) Simple interest. (3 marks)
- ii) Compound interest. (3 marks)
- iii) Calculate the difference between simple and compound interest. (2 marks)

b) A school bus was valued at sh 6,000,000 in January 2000. Each year, its value decreased by 12.5% of its value at the beginning of the year. Find the value of the bus in January 2005, giving your answer correct to 4 significant figures (2 marks)

20.a) Complete the table below for the function $y = -2x^2 - 7x + 4$ (2marks)

x	-5	-4	-3	-2	-1	0	1
$-2x^2$	-50	-32		-8		0	
$-7x$	35		21			0	
$+4$	4	4	4	4	4	4	4
$y = -2x^2 - 7x + 4$	-11		7			4	-5

(b) Draw the graph of $y = -2x^2 - 7x + 4$ for $-5 \leq x \leq 1$. (3marks)

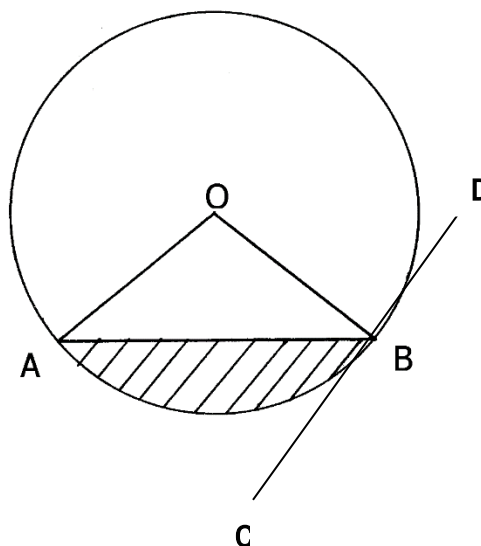
(c) Use your graph to solve

(i) $-2x^2 - 7x + 4 = 0$ (1mark)

(ii) $-2x^2 - 4x - 2 = 0$ (2marks)

(iii) $-2x^2 - 7x + 2 = 0$ (2marks)

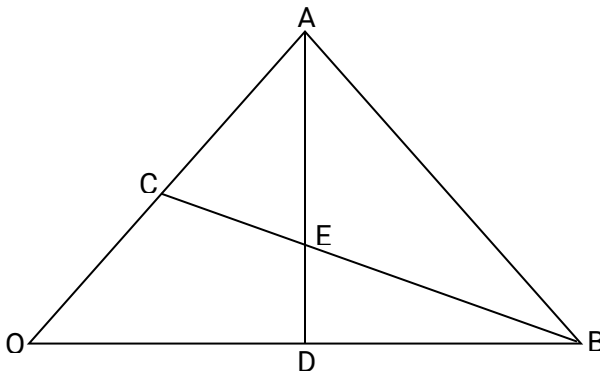
21. In the figure below, O is the centre of the circle of radius 3cm and AB is a chord such that its shortest distance from O is 1cm. CBD is a tangent to the circle at B.



Calculate:

- a) The length of the chord AB. (2marks)
- b) Angle AOB (2marks)
- c) The area of the minor sector OAB. (2marks)
- d) The area of the shaded segment. (2marks)
- e) The angle ABC. (2marks)

22. The diagram below shows triangle OAB. $OA = a$, $OB = b$. C divides OA in the ratio 1:2 and D divides OB in the ratio 2:3 while AD and BC meet at E.

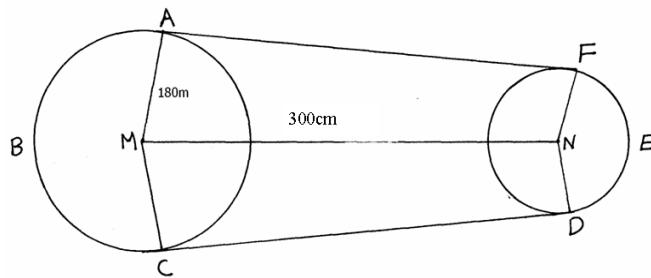


- a) Find the following vectors in terms of a and b
 - (i) OC . (1 mark)
 - (ii) CB . (1 mark)
 - b) Given that $CE = mCB$ and $AE = nAD$, where m and n are scalars, find:
 - (i) Two expressions for OE . (3 marks)
 - (ii) The values of m and n . (3 marks)
 - (iii) OE in terms of a and b only. (1 mark)
 - (iv) The ratio $CE:EB$. (1 mark)
- 23.(a) Three quantities P , Q and R are such that P varies directly as the square of Q and inversely as the square root of R .
- i) Given that Q increases by 5% and R decreases by 36%, find the percentage change in P . (5 marks)

ii) If $P = 6$ when $Q = 12$ and $R = 25$, find the value of P when $Q = 15$ and $R = 81$. (3 marks)

b) After the changes in (a) i) above, what would be the value of P found in (a) (ii) above. (2 marks)

24.a) The figure below shows a pulley system where a conveyer belt is tied round the two wheels. The radius of the larger wheel is 180cm and the distance between the centres of the wheels is 300cm and angle $AMC = 140^\circ$.



Determine

(a) the length of AF . (2marks)

(b) the length of the arc FED . (4marks)

(c) the length of the arc ABC . (2marks)

(d) the total length of the conveyer belt. (2marks)

EXAM 6 PAPER 1

Section I (50 marks)

Answer all the questions in this section

1. Evaluate: $\frac{\frac{4}{11} \text{ of } \left(\frac{3}{4} - \frac{1}{20}\right)}{\left(3 + \frac{1}{3}\right) \div \left(1 + \frac{1}{10}\right)}$ without using calculator or mathematical tables leaving your answer as a simplified fraction. (2 marks)

2. Find the equation of a line perpendicular to $2x + 4y = 8$, which cross the line at its y - Intercept. (3 marks)

3. Simplify completely: $\frac{3x^2 - 1}{x^2 - 1} - \frac{2x + 1}{x + 1}$ (3 marks)

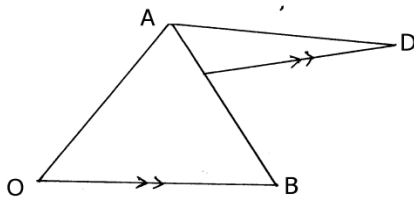
4. Find the integral values of x for which:
 $-6 \leq 3x + 3$
 $14 - 3x > 2$ (3 marks)

5. Use tables of square, cubes roots and reciprocals to find the value of x if
 $x = \sqrt[3]{\frac{1}{0.2365} + \frac{2}{(2.6228)^2}}$ (3 marks)

6. One interior angle of a certain polygon is 84° . If each of the other angles is 147° , how many sides does this polygon have? (3 marks)

7. Simplify $\sqrt{\frac{ax^3}{b} \div \frac{b^{-3}}{a^3x^5}}$ (3 marks)

8. In the figure below $\mathbf{OA} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{OB} = 8\mathbf{i}$. C is a point on AB such that AC: CB = 3: 2, and D is a point such that $\mathbf{OB} \parallel \mathbf{CD}$ and $2\mathbf{OB} = \mathbf{CD}$.



Determine the vector $\mathbf{DA}$ in terms of $\mathbf{i}$ and $\mathbf{j}$. (3 marks)

9. An airbus left town N at 1945hrs and arrived in town L at 0320hrs. It stayed for $1\frac{1}{2}$ hours for rest and refreshment of passengers and crew. It then headed for town W and took $1\frac{1}{2}$ hours.

(a) How long did the journey from town N to town L take in hours and minutes? (2 marks)

(b) At what time did it arrive in town W? (2 marks)

10. Mzee is three times as old as his son now. If ten years ago the sum of their ages was 44, how old was Mzee when his son was born? (3 marks)

11. Given that $\frac{2x+3y}{3x+4y} = 5$, find the ratio $y : x$. (3 marks)

12. A cylinder of radius 14cm contains water. A metal solid hemisphere of radius 6.8cm is submerged into the water in the cylinder. Find the change in height of the water in the cylinder. (4 marks)

13. An Italian tourist, on arrival in Kenya, converted 6000 Euros into Kenya shillings. During her three day's stay the country, she spent Kshs 260,000. She converted the remaining amount into US dollars. Using the exchange rates table below calculate how many US dollars she got?

	Buying	Selling
1 US dollar \$	96.20	96.90
1 Euro C	112.32	112.83

(3 marks)

14. Find the area in hectors of a field book measurement is recorded in metres as follows.

	D	
	170	
To E 50140	110	60 to C
To F 80100	30	50 to B
	A	

(3 marks)

15. The perimeter of a triangular field is 120 m. Two of the sides are 21m and 40m. Calculate the largest angle of the field hence find the area of the field. (4 marks)

16. Given that $\log 4 = 0.6021$ and $\log 6 = 0.7782$, without using mathematical tables or a calculation, evaluate $\log 0.096$. (3 marks)

SECTION II- 50 MARKS

Answer any five questions in this section

17. A Kazi Kwa Vijana youth group decided to raise sh.126,000 towards a quail farming project. Each youth was to contribute an equal amount of money. Before the contributions, 7 of them pulled out the group resulting in the remaining members to pay more.

(a) If x stands for the original number of members, show that the increase in contribution was $\frac{882,000}{x(x-7)}$

(3marks)

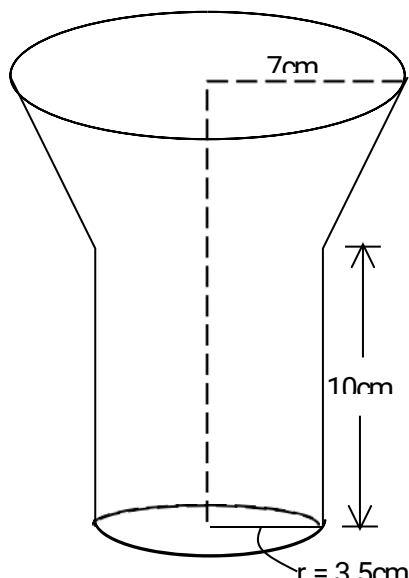
(b) If the increase in contribution per youth was sh.200, find the number of youths that were originally involved in the project.

(4marks)

(c) Calculate percentage increase in the contribution per youth caused by the withdrawal of some women to one decimal place.

(3marks)

18. The diagram below represents a model of a pillar. The radii of the top and the base are 7cm and 3.5cm respectively. The height of the cylindrical part is 10cm while the height of the whole pillar is 15cm.



(a) Calculate the volume of the model in cm^3 .

(6 marks)

(b) Calculate the mass of the material used to construct the pillar given that the actual height of the whole pillar is 60m and the density of the material used is 0.832g/cm^3 . (Give your answer in tonnes).

(4 marks)

19. A bus left N at 7.00a.m and travelled towards town E at an average speed of 80 km/h. At 7.45a.m a car left town E towards town N at an average speed of 120 km/h. The distance between towns N and E is 300km. Calculate:

(a) The time the bus arrived at town E.

(2 marks)

(b) The time of the day the two met.

(4 marks)

c) The distance from Nairobi where the two met. (2 marks)

d) The distance of the bus from town E when the car arrived at town N. (2 marks)

20. In a physics test student scored the following marks.

72 50 43 58 62 49 69 60 84 62 55 89 67 92 81 75 63 77 95 65 54

35 45 73 41 56 50 36 49 58 61 85 38 64 76 78 51 43 72 37 62 55

(a) Using a class width of 10 and 35-44 as the first class, make a frequency table of the grouped data. (5marks)

(b) Estimate

(i) The mean (2marks)

(ii) The median (3marks)

21. (a) Find the equation of a straight line passing through the points (3, 2) and (-3, 6) giving your answer in the form $\frac{x}{a} + \frac{y}{b} = 1$, where a and b are constants. (4marks)

(b) State the coordinates of point A and B, at which the line in (a) above crosses the x-axis and y-axis respectively. (2marks)

(c) Using the information in (a) and (b) above, find the area of triangle AOB, where O is the origin (2marks)

(d) Find the acute angle the line in (a) above makes with the axis. (2marks)

22. A flag post 12m long is fixed on top of a tower. From a point on horizontal ground, the angles of elevation of the top and bottom of the flag post are 46° and 33° respectively.

Calculate

(a) The horizontal distance from the point on the ground to the base of the tower (4marks)

(b) The total height of the tower and the flag. (2marks)

c) The shortest distances from the point on the ground to

i) The top of the flag post. (2marks)

ii) The top of the tower. (2marks)

23. The vertices of triangle PQR are P (0, 0), Q (6, 0) and R (2, 4).

a) Plot these points on the grid provided below. (1 mark)

b) Triangle $P^1Q^1R^1$ is the image of a triangle PQR under an enlargement scale factor $\frac{1}{2}$ and centre (2, 2). Write down the coordinates of triangle $P^1Q^1R^1$ and plot on the same grid.

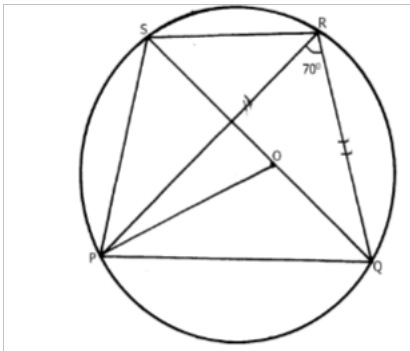
(2 marks)

c) Draw triangle $P^{11}Q^{11}R^{11}$ the image of triangle $P^1Q^1R^1$ under a positive quarter turn about the point (1, 1).
(2 marks)

d) Draw a triangle $P^{111}Q^{111}R^{111}$ the image of triangle $P^1Q^1R^1$ under reflection in the line $y = 1$.
(3 marks)

e) Describe fully a single transformation which maps triangle $P^{11}Q^{11}R^{11}$ onto triangle $P^1Q^1R^1$.
(2 marks)

24. The figure below shows a circle centre O. PQRS is a cyclic quadrilateral and QOS is a straight line.



Giving reasons for your answers, find the value of:

- | | | |
|----|-----------------------|-----------|
| a) | $\angle PRS$. | (2 marks) |
| b) | $\angle POQ$. | (2 marks) |
| c) | $\angle RPS$. | (2 marks) |
| d) | $\angle PSR$. | (2 marks) |
| e) | Reflex $\angle POS$. | (2 marks) |

PAPER 2

Section I (50 marks)

Answer all the questions in this section

1. Use logarithms to evaluate correct to 4 significant figures

$$\left(\frac{\sin 54.5}{\tan 24.8 \times \cos 78} \right)^{1/2}$$

(4marks)

2. A blend of juice is made from pineapple and passion. The cost of two litres of pineapple is Kshs 120 and three litres of passion is Kshs 270. In what ratio should the juices be mixed such that by selling the mixture at Kshs 84 per litre a profit of 20% is realized?
(3 marks)

3. A ball allowed to drop from a height of 16m on to a floor rebounds to $\frac{3}{4}$ of its previous height. Find the

total heights the ball will have rose when it hits the ground for the tenth time correct to four significant figures 1. (2 marks)

4. Solve the equation

$$\sqrt{y} = \frac{2}{\sqrt{y}} + 1.$$

(3 marks)

5. Use completing the square method to solve for x in the equation: $3x^2 + 18x + 15 = 0$. (3 marks)

6. A shamba is in the shape of a parallelogram with the lengths of the adjacent sides being 12cm and 15cm. If the area of the parallelogram is 72cm^2 , find the angle between these two sides. (3 marks)

7. A quantity y varies partly as x^2 and partly as x . When $y = 6$, $x = 1$ when $y = 30$, $x = 3$. Find y when $x = -3$. (3 marks)

8. Two taps A and B together, can fill a water tank in 6 minutes. Tap A alone takes 5 minutes longer to fill the tank than tap B alone. How many minutes does it take tap B alone to fill the tank? (3 marks)

9. Hisabati originally worked out the mean mark of her forty pupils to be forty one. After confirmation of the marks with her pupils, she added some marks to Tim, Chob and Chel in the ratio 5:2:3 respectively. If the new mean mark for the class increased by 1.5, find how many marks Chel was added than Chob. (3 marks)

10. (a) Write down the first five terms of the expansion of $\left(1 - \frac{x}{3}\right)^5$. (2 marks)

- b) Using the first three terms of the expansion. Find the values of $(1.01)^5$ to 4 decimal places. (2marks)

11. A school bus was valued at shs 6,000,000 in January 2010. It depreciated by 12.5% of its value at the beginning of the year. Find the value of the bus in January 2015, giving your answer correct to 4 significant figures. (3 marks)

12. The shadow of a vertical pole is observed on level ground to be 20m long when the angle of elevation of the sun is 30° . Calculate the reduction in the length of the shadow of the pole when the angle of elevation the sun becomes 80° . (3 marks)

13. A rectangle measures 8.6 cm and 4.8 cm. Find the limits within which the area of the rectangle lies hence find the percentage error in the area. (3 marks)

14. Solve the equation: $\log_2(2 + 3x) + 3 \log_2 2 = 2 + \log_2(2x + 6)$. (3 marks)

15. The points with coordinates (5, 5) and (-3, 1) are the ends of the diameter of a circle centre A.
Determine:
- (a) The coordinates of A. (1mark)
- (b) The equation of the circle, expressing it in form $x^2 + y^2 + ax + by + c = 0$ Where a, b, and c are constants. (3 marks)
16. Make p the subject of the formula: $x = \left(\frac{1}{mp^3} - A^2 \right) B$. (3 marks)

SECTION B: (50 MARKS)

Answer any FIVE questions from this section

17. The table below shows income tax rates for the year 2014.

Income in Kenya pounds per month	Rate in Kshs per pound
1 – 484	2
485 – 940	3
941 – 1396	4
1397 – 1852	5
Over 1852	6

In the year of income 2014, the tax on Bushuru's monthly taxable income was Kshs 10,880.50. Bushuru was entitled to a tax relief of Kshs 1,156.

Find

- i) Bushuru's gross monthly tax. (2marks)
- ii) Bushuru's taxable income per month in shillings. (6marks)
- b) Apart from basic salary, Bushuru earned a house allowance of Kshs 12,000 a medical allowance of Kshs 3,060 and a hardship allowance of Kshs 4,635. Find his basic salary per month. (2marks)
18. A number is selected at random from 2, 3, 5, 7, 9, and 11 and paired with another number selected from 4, 6, 8, 10, 12, and 14.
- (a) Construct a table showing how the numbers are paired. (2 marks)
- (b) Find the probability that the sum of the selected numbers is even. (2 marks)
- (c) Find the probability that the sum is odd and prime. (3 marks)

- (d) Find the probability that the sum is greater than the greatest number that can be selected.

(3 marks)

19. Two variables A and B are believed to be related by a law of the form $A = mn^B + 1.1$ where m and n are constants. The table below shows corresponding values of A and B as obtained from an experiment.

By drawing a suitable straight line graph estimate the values of m and n.

(10 marks)

20. Matrix **P** is given by $\begin{pmatrix} 4 & 7 \\ 5 & 8 \end{pmatrix}$.

- (a) Find P^{-1} (2marks)

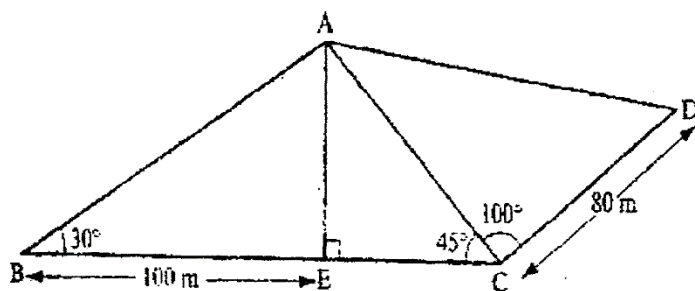
- (b) Two traders Mwanzo and Mwisho purchased beans at Kshs x per bag and maize at Kshs y per bag. Mwanzo purchased 8 bags of beans and 14 bags of maize for Kshs 47,600. Mwisho purchased 10 bags of beans and 16 bags of maize for sh. 57,400.

- (i) Form a matrix equation to represent the information above. (2marks)

- (ii) Use the matrix P^{-1} to find the prices of one bag of each item (3marks)

- (c) The price of bean later went up by 5% and that of maize remained constant. Mwanzo bought the same quantity of beans but spent the same total amount of money as before on the two items. State the new ratio of beans to maize. (3marks)

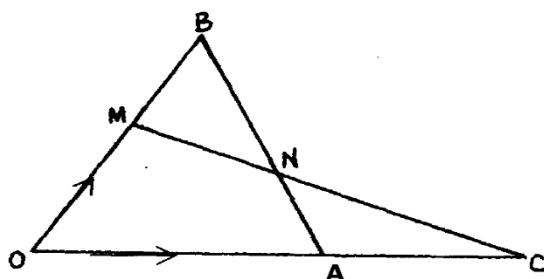
21. The figure below represents a quadrilateral piece of land ABCD divided into three triangular plots. The lengths **BE** and **CD** are 100m and 80m respectively. Angle $\angle ABE = 30^\circ$, $\angle ACE = 45^\circ$ and $\angle ACD = 100^\circ$



Find to four significant figures:

- (a) The length of AE (3marks)
- (b) The length of AD (3marks)
- (c) The perimeter of the piece of land (4marks)

22. In the triangle OAB below, $OA = a$, $OB = b$ and $OC = \frac{3}{2}OA$. M divides OB in the ratio 3:2



a) Express in terms of **a** and **b** only, the vector

i) $\overrightarrow{AB}$ (1 mark)

ii) $\overrightarrow{MC}$ (1 mark)

b) Given that $\overrightarrow{MN} = h\overrightarrow{MC}$ and $\overrightarrow{BN} = k\overrightarrow{BA}$, express vector $\overrightarrow{MN}$ in two different ways and hence, find the value of h and k . (6 marks)

c) Show that the points M , N and C are collinear. (2 marks)

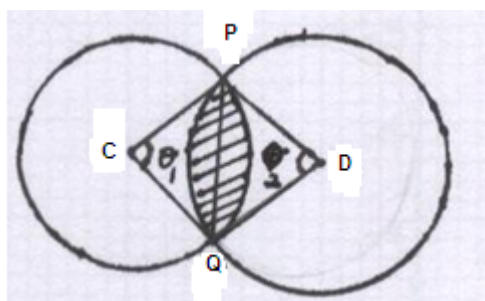
23.(a) The fifth term of an arithmetic progression is 11 and the twenty fifth term is 51. Find the first term and common difference of the progression. (3 marks)

b) The second and fifth terms of a geometric progression are 16 and 2 respectively. Determine the common ratio and the first term (3marks)

c) Find the sum of the following GP. (4 marks)

$$2 + 10 + 50 + \dots + 1250$$

24. The figure below shows two intersecting circles of centres C and D radii 16cm and 20cm respectively. The two circles subtend angles θ_1 and θ_2 at their centres respectively and intersect at P and Q as shown.



a) Given that the area of triangle PCQ is 80.14cm^2 , calculate the size of

i) The angle marked θ_1 . (2marks)

ii) The angle marked θ_2 . (3marks)

b) Calculate the area of the shaded region.

(5marks)

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